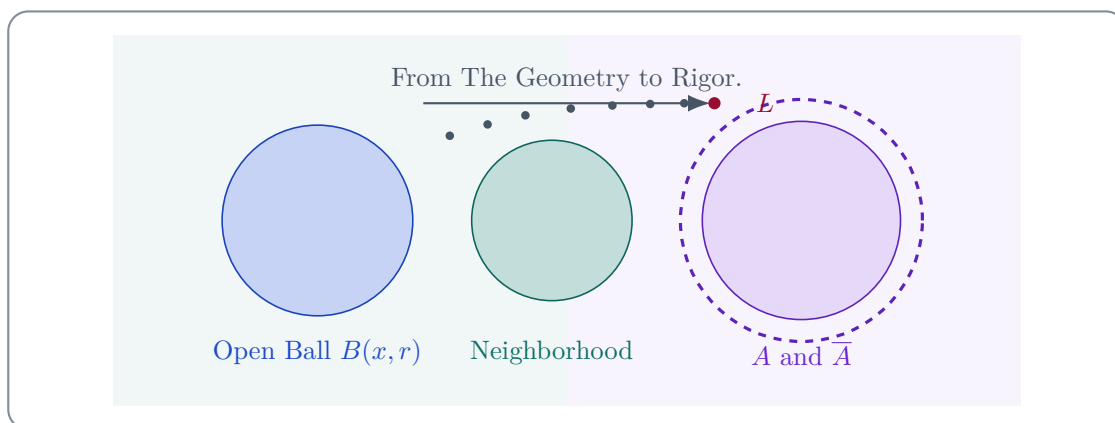


Visual Foundations: Open Sets, Boundary, Connectedness, and Orbits

A Visual and Problem-driven Introduction

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Rigor is The Art of Making Intuition Unavoidable.



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Preface

This handout are prepared to help you on the core ideas of analysis and topology through a single guiding principle: **think in neighborhoods**. Rather than treating definitions as formalities, we use them as practical tools for proving statements and for reading diagrams correctly. The first aim is to build a reliable geometric–logical vocabulary around **open sets** in $\mathbb{R}^n$. Openness is introduced via the ball criterion: a point belongs to an open set precisely when a small Euclidean ball around it stays entirely inside the set. This local viewpoint immediately clarifies common confusions, especially the role of the **boundary**: an open set never contains its boundary points, yet it can have a rich and nonempty boundary as a separate set. Throughout, diagrams are used not as decoration but as “proof-guides” that mirror the logical structure of the definitions. Next, we developed **connectedness** and **path-connectedness**, beginning with concrete separations in $\mathbb{R}^n$ and then moving to the key theorem:

$$\text{open and connected in } \mathbb{R}^n \implies \text{path-connected.}$$

The proof is presented in a standard rigorous form (via the reachable set A), but we also include a visual roadmap that explains *why* the argument works: openness creates small balls that allow local “motion,” and connectedness prevents the space from splitting into two disjoint relatively open pieces. A recurring theme is that not every geometric intuition is safe without hypotheses. To highlight this, we briefly step beyond Euclidean regions and include the classical **topologist’s sine curve**, a set that is connected but not path-connected. This example is included to emphasize the special role of *openness* in $\mathbb{R}^n$: regions behave better precisely because local ball neighborhoods give enough room to build paths.

Finally, we introduce **orbits under iteration** as a motivating bridge from “repeated application” to long-term geometric behavior. Even a simple rule $f(x)$ generates rich phenomena when iterated: bounded orbits, periodic cycles, escaping behavior, and qualitatively complicated motion. The quadratic family $f_c(z) = z^2 + c$ serves as a concrete laboratory in the complex plane, and the escape-radius viewpoint is presented both algebraically and visually.

Each section follows a consistent learning rhythm:

- Boxed definitions designed for quick recall,
- Carefully written proofs (with the key idea made explicit),
- Diagrams that match the logic of the proof,
- Solved examples that model problem-solving style,
- Exercises that reinforce technique and intuition.

This note assume comfort with basic real analysis and Euclidean geometry, and they offer a gentle first contact with topological language (open sets, boundaries, connectedness, paths) without requiring a full course in general topology. The goal is to make you confident in moving between pictures and proofs:

Draw the idea, state it in neighborhoods, and prove it cleanly.

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1 Boundary: Where a Set Meets Its Surroundings

Idea / Intuition. Geometric Meaning

A point lies on the boundary of A if you can approach it both from *inside* A and from *outside* A . Formally: every neighborhood intersects both A and $X \setminus A$.

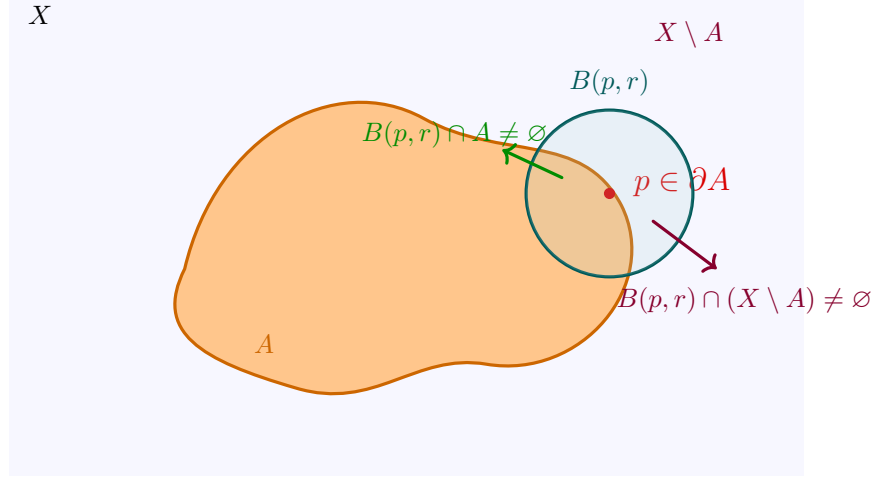


Figure 1: A boundary point $p \in \partial A$: every neighborhood meets both A and $X \setminus A$.

In the figure, the orange region represents the set A and the surrounding region represents its complement $X \setminus A$. The point p is drawn on the edge of the orange region. The teal circle is a typical neighborhood $B(p, r)$. Because p lies on the boundary, *no matter how small* we choose $r > 0$, the neighborhood $B(p, r)$ contains points of A (shown by the arrow pointing into the orange region) and also contains points of $X \setminus A$ (shown by the arrow pointing into the outside region). This is precisely what distinguishes boundary points from interior points (whose small neighborhoods lie entirely in A) and exterior points (whose small neighborhoods lie entirely in $X \setminus A$).

1.1 Definitions and Fundamental Equivalences

Definition. Boundary Point and Boundary

Let (X, τ) be a topological space and $A \subseteq X$. A point $x \in X$ is a *boundary point* of A if for every open set $U \in \tau$ with $x \in U$,

$$U \cap A \neq \emptyset \quad \text{and} \quad U \cap (X \setminus A) \neq \emptyset.$$

The set of all boundary points is called the *boundary* of A and is denoted by ∂A .

Definition. Closure and Interior (Neighborhood Form)

Let $A \subseteq X$.

- $x \in \bar{A}$ iff every neighborhood of x meets A .
- $x \in A^\circ$ iff there exists a neighborhood U of x with $U \subseteq A$.

Theorem. Boundary via Closure/Interior

For any $A \subseteq X$ in a topological space (X, τ) ,

$$\partial A = \overline{A} \setminus A^\circ \quad \text{and} \quad \partial A = \overline{A} \cap \overline{(X \setminus A)}.$$

Recall the Definitions (Neighborhood Form:) Let $x \in X$.

- $x \in \partial A$ means: for every open set $U \in \tau$ with $x \in U$,

$$U \cap A \neq \emptyset \quad \text{and} \quad U \cap (X \setminus A) \neq \emptyset.$$

- $x \in \overline{A}$ means: for every open set $U \in \tau$ with $x \in U$,

$$U \cap A \neq \emptyset.$$

- $x \in A^\circ$ means: there exists an open set $U \in \tau$ such that

$$x \in U \subseteq A.$$

Equivalently, $x \notin A^\circ$ means: for every open set $U \in \tau$ with $x \in U$,

$$U \not\subseteq A.$$

Step 1: Prove $\partial A = \overline{A} \setminus A^\circ$.

($\subseteq$). Assume $x \in \partial A$. We show $x \in \overline{A}$ and $x \notin A^\circ$.

(i) Show $x \in \overline{A}$. Let U be any open set with $x \in U$. Since $x \in \partial A$, by definition we have $U \cap A \neq \emptyset$. This holds for *every* such open U , hence $x \in \overline{A}$.

(ii) Show $x \notin A^\circ$. Suppose, for contradiction, that $x \in A^\circ$. Then by definition of interior, there exists an open set V such that

$$x \in V \subseteq A.$$

But then $V \cap (X \setminus A) = \emptyset$ (because $V \subseteq A$). This contradicts the boundary condition for $x \in \partial A$, which requires *every* open neighborhood of x (in particular V) to meet $X \setminus A$. Therefore $x \notin A^\circ$.

Combining (i) and (ii), we have $x \in \overline{A} \setminus A^\circ$.

($\supseteq$). Assume $x \in \overline{A} \setminus A^\circ$. We show $x \in \partial A$.

Let U be an arbitrary open set with $x \in U$. We must prove two statements:

$$U \cap A \neq \emptyset \quad \text{and} \quad U \cap (X \setminus A) \neq \emptyset.$$

(i) Show $U \cap A \neq \emptyset$. Since $x \in \overline{A}$, by definition every open neighborhood of x meets A . In particular, $U \cap A \neq \emptyset$.

(ii) Show $U \cap (X \setminus A) \neq \emptyset$. Because $x \notin A^\circ$, there does *not* exist an open set containing x that lies entirely in A . Equivalently, for every open set U with $x \in U$, we have $U \not\subseteq A$. Apply this to our chosen U : since $U \not\subseteq A$, there exists a point $u \in U$ with $u \notin A$. But $u \notin A$ means $u \in X \setminus A$, so $u \in U \cap (X \setminus A)$. Hence $U \cap (X \setminus A) \neq \emptyset$.

Since U was arbitrary, both intersections are nonempty for every neighborhood of x , so by definition $x \in \partial A$. Therefore, $\partial A = \overline{A} \setminus A^\circ$.

Step 2: Deduce $\partial A = \overline{A} \cap \overline{(X \setminus A)}$.

We show both inclusions.

($\subseteq$). If $x \in \partial A$, then by definition every neighborhood U of x meets A and also meets $X \setminus A$. Hence $x \in \overline{A}$ and $x \in \overline{(X \setminus A)}$, so $x \in \overline{A} \cap \overline{(X \setminus A)}$.

($\supseteq$). If $x \in \overline{A} \cap \overline{(X \setminus A)}$, then every neighborhood U of x meets A (since $x \in \overline{A}$) and meets $X \setminus A$ (since $x \in \overline{(X \setminus A)}$). Thus x satisfies the boundary condition, so $x \in \partial A$.

Hence $\partial A = \overline{A} \cap \overline{(X \setminus A)}$. ■

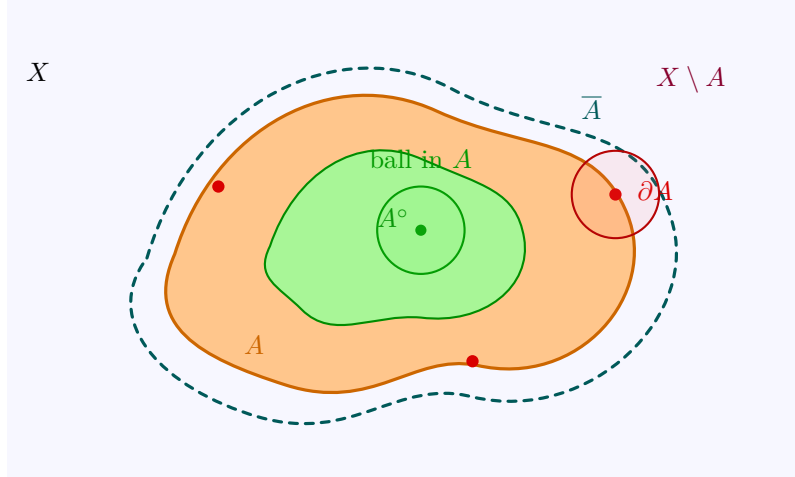


Figure 2: Visual intuition for $\partial A = \overline{A} \setminus A^\circ$ and $\partial A = \overline{A} \cap \overline{(X \setminus A)}$.

How the Figure Encodes the Boundary Identities: In the picture, the orange region is the set $A \subseteq X$. The green region inside it represents the interior A° , i.e. points having a small open ball entirely contained in A (illustrated by the green ball). The red dots lie on the boundary ∂A . Around a boundary point $p \in \partial A$, the red ball indicates a typical neighborhood $B(p, r)$. Such a neighborhood necessarily intersects both A and $X \setminus A$; hence every boundary point belongs to $\overline{A}$ (because every neighborhood meets A) but does *not* belong to A° (because no neighborhood can be contained in A). This visually explains

$$\partial A = \overline{A} \setminus A^\circ.$$

Equivalently, a point is on the boundary precisely when it is simultaneously adherent to A and to its complement: every neighborhood meets A and every neighborhood meets $X \setminus A$. This is represented by the boundary points sitting exactly where both “sides” are unavoidable, matching the identity

$$\partial A = \overline{A} \cap \overline{(X \setminus A)}.$$

Proposition. Basic Properties of The Boundary

For any $A \subseteq X$ in a topological space (X, τ) :

- (a) ∂A is closed.
- (b) $\partial A = \partial(X \setminus A)$.
- (c) $\partial A = \emptyset$ iff A is clopen (both open and closed).

Proof: We repeatedly use the identities proved earlier:

$$\partial A = \overline{A} \setminus A^\circ \quad \text{and} \quad \partial A = \overline{A} \cap \overline{(X \setminus A)}.$$

(a) ∂A is closed.

Method 1 (Via Closure-Intersection). Using

$$\partial A = \overline{A} \cap \overline{(X \setminus A)},$$

note that $\overline{A}$ is closed and $\overline{(X \setminus A)}$ is closed (closures are always closed). An intersection of closed sets is closed, hence ∂A is closed.

(Optional check: Why Closures are Closed). By definition, $\overline{B}$ is the smallest closed set containing B ; in particular $\overline{B}$ itself is closed.

(b) $\partial A = \partial(X \setminus A)$.

Apply the intersection formula to the complement:

$$\partial(X \setminus A) = \overline{(X \setminus A)} \cap \overline{X \setminus (X \setminus A)}.$$

But $X \setminus (X \setminus A) = A$, so this becomes

$$\partial(X \setminus A) = \overline{(X \setminus A)} \cap \overline{A}.$$

Since intersection is commutative,

$$\overline{(X \setminus A)} \cap \overline{A} = \overline{A} \cap \overline{(X \setminus A)} = \partial A.$$

Hence $\partial A = \partial(X \setminus A)$.

(c) $\partial A = \emptyset$ iff A is clopen.

($\Rightarrow$) Assume $\partial A = \emptyset$. Show A is open and closed.

From $\partial A = \overline{A} \setminus A^\circ$ we get

$$\emptyset = \overline{A} \setminus A^\circ.$$

This means there are *no* points in $\overline{A}$ that are outside A° , i.e.

$$\overline{A} \subseteq A^\circ.$$

On the other hand, by general facts about interior and closure we always have

$$A^\circ \subseteq A \subseteq \overline{A}.$$

Combining these inclusions yields a chain

$$A^\circ \subseteq A \subseteq \bar{A} \subseteq A^\circ,$$

so all sets are equal:

$$A^\circ = A = \bar{A}.$$

Now:

- $A = A^\circ$ implies A is open (a set equals its interior iff it is open).
- $A = \bar{A}$ implies A is closed (a set equals its closure iff it is closed).

Therefore A is clopen.

($\Leftarrow$) **Assume A is clopen. Show $\partial A = \emptyset$.**

If A is open, then $A^\circ = A$. If A is closed, then $\bar{A} = A$. Substitute into $\partial A = \bar{A} \setminus A^\circ$:

$$\partial A = \bar{A} \setminus A^\circ = A \setminus A = \emptyset.$$

Hence $\partial A = \emptyset$.

This proves the equivalence in (c), completing the proposition. ■

Proposition. Neighborhood Test (Boundary vs. Interior vs. Exterior)

Let $A \subseteq \mathbb{R}^n$ and $x \in \mathbb{R}^n$. Then:

- (a) **Interior Test:** $x \in A^\circ$ iff $\exists r > 0$ such that $B(x, r) \subseteq A$.
- (b) **Exterior Test:** $x \notin \bar{A}$ iff $\exists r > 0$ such that $B(x, r) \subseteq \mathbb{R}^n \setminus A$.
- (c) **Boundary Test:** $x \in \partial A$ iff $\forall r > 0$,

$$B(x, r) \cap A \neq \emptyset \quad \text{and} \quad B(x, r) \cap (\mathbb{R}^n \setminus A) \neq \emptyset.$$

Practical Numeric Check (Intuition): choose radii $r_k \downarrow 0$ (e.g. $r_k = 10^{-k}$) and test whether each ball $B(x, r_k)$ hits *both* A and its complement. If it does for all tested k , the point behaves like a boundary point (for an exact proof, use the definition/closure/interior identities).

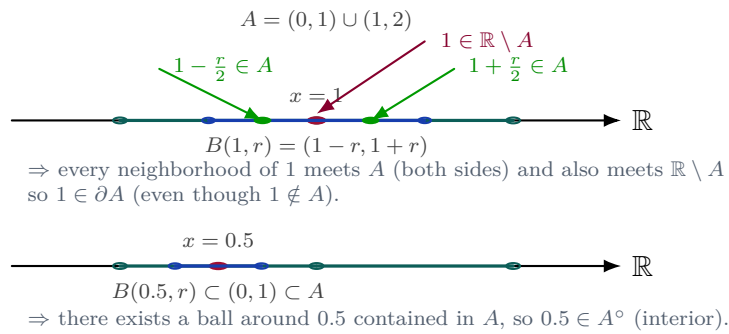


Figure 3: Numeric neighborhood test for $A = (0, 1) \cup (1, 2)$: 1 is a boundary point (but not in A), while 0.5 is interior.

1.2 Solved Examples

Solved Example. Open Interval and Boundary in $\mathbb{R}$

Compute $\partial(0, 1)$ in $\mathbb{R}$.

Solution: Recall the identity (proved earlier)

$$\partial A = \overline{A} \setminus A^\circ.$$

So we compute $\overline{(0, 1)}$ and $(0, 1)^\circ$.

Step 1: Compute the Interior: Since $(0, 1)$ is an open interval in $\mathbb{R}$, every point $x \in (0, 1)$ has some radius $r > 0$ such that

$$(x - r, x + r) \subset (0, 1).$$

Hence $(0, 1)$ is open, and therefore

$$(0, 1)^\circ = (0, 1).$$

Step 2: Compute the Closure: A point $x \in \mathbb{R}$ lies in $\overline{(0, 1)}$ iff every open interval $(x - r, x + r)$ meets $(0, 1)$. We show that

$$\overline{(0, 1)} = [0, 1].$$

(i) Show $[0, 1] \subseteq \overline{(0, 1)}$. Let $x \in (0, 1)$. Then $x \in (0, 1)$, so every neighborhood of x meets $(0, 1)$, hence $x \in \overline{(0, 1)}$. Now consider the endpoints:

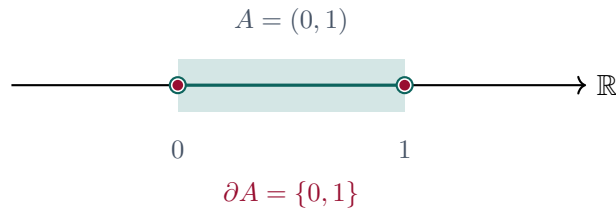
- For $x = 0$: given any $r > 0$, the interval $(-r, r)$ contains points in $(0, 1)$ (e.g. $r/2$ if $r < 2$, or $1/2$ if $r \geq 2$). So every neighborhood of 0 intersects $(0, 1)$, hence $0 \in \overline{(0, 1)}$.
- For $x = 1$: given any $r > 0$, the interval $(1 - r, 1 + r)$ contains points in $(0, 1)$ (e.g. $1 - r/2$ if $r < 2$, or $1/2$ if $r \geq 2$). So $1 \in \overline{(0, 1)}$.

Thus $[0, 1] \subseteq \overline{(0, 1)}$.

(ii) Show $\overline{(0, 1)} \subseteq [0, 1]$. Take $x \notin [0, 1]$. Then either $x < 0$ or $x > 1$.

- If $x < 0$, set $r = \frac{-x}{2} > 0$. Then $(x - r, x + r) \subset (-\infty, 0)$, so it does not meet $(0, 1)$. Hence $x \notin \overline{(0, 1)}$.
- If $x > 1$, set $r = \frac{x-1}{2} > 0$. Then $(x - r, x + r) \subset (1, \infty)$, so it does not meet $(0, 1)$. Hence $x \notin \overline{(0, 1)}$.

So no point outside $[0, 1]$ is in the closure, giving $\overline{(0, 1)} \subseteq [0, 1]$. Combining (i) and (ii), we conclude $\overline{(0, 1)} = [0, 1]$.



Step 3: Compute the Boundary. Now apply $\partial A = \overline{A} \setminus A^\circ$:

$$\partial(0, 1) = \overline{(0, 1)} \setminus (0, 1)^\circ = [0, 1] \setminus (0, 1) = \{0, 1\}.$$

Solved Example. Boundary of a Half-open Interval

Compute $\partial(0, 1]$ in $\mathbb{R}$.

Solution: Let $A = (0, 1]$.

Step 1: Compute $\overline{A}$: We claim $\overline{A} = [0, 1]$.

- Since $A \subseteq [0, 1]$ and $[0, 1]$ is closed, we have $\overline{A} \subseteq [0, 1]$.
- Conversely, take any $x \in [0, 1]$. For every $\varepsilon > 0$, the interval $(x - \varepsilon, x + \varepsilon)$ meets A :
 - If $x \in (0, 1]$, then $x \in A$ already.
 - If $x = 0$, then $(0, \varepsilon) \subset A$ for $\varepsilon \leq 1$.

So every neighborhood of x meets A , hence $x \in \overline{A}$.

Thus $\overline{A} = [0, 1]$.

Step 2: Compute A° : We claim $A^\circ = (0, 1)$.

- If $x \in (0, 1)$, choose $r = \min\{x, 1 - x\} > 0$. Then $(x - r, x + r) \subset (0, 1) \subset A$, so $x \in A^\circ$.
- If $x = 1$, then every interval $(1 - r, 1 + r)$ contains points > 1 , which are not in A , so no neighborhood of 1 is contained in A . Hence $1 \notin A^\circ$.

Therefore $A^\circ = (0, 1)$.

Step 3: Boundary: Using $\partial A = \overline{A} \setminus A^\circ$,

$$\partial(0, 1] = [0, 1] \setminus (0, 1) = \{0, 1\}.$$

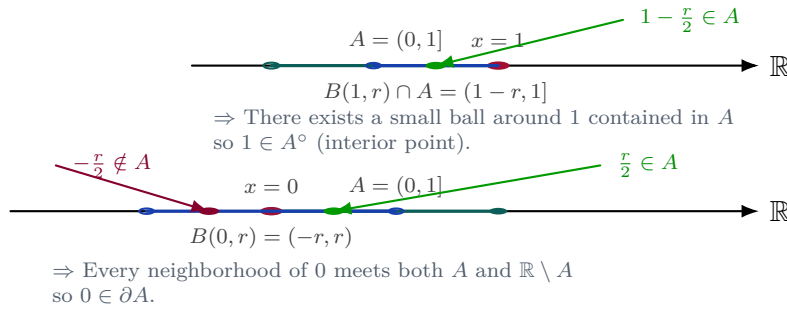


Figure 4: Numeric neighborhood test for $A = (0, 1]$: 1 is an interior point, while 0 is a boundary point.

Numeric Check (Intuition): Pick $x = 1$. For any $r > 0$, the point $1 + \frac{r}{2} \in (1 - r, 1 + r)$ lies outside A , so 1 cannot be interior. Pick $x = 0$. For any $r > 0$, the point $\frac{r}{2} \in (0, r) \subset (0, 1] \subset A$ and also $-\frac{r}{2} < 0$ lies outside A , so every neighborhood of 0 hits both sides $\Rightarrow 0$ is boundary.

Solved Example. Boundary of An Open Interval

Compute $\partial(0, \infty)$ in $\mathbb{R}$.

Solution: Let $A = (0, \infty)$.

Step 1: $\overline{A} = [0, \infty)$. Clearly $A \subseteq [0, \infty)$ and $[0, \infty)$ is closed, so $\overline{A} \subseteq [0, \infty)$. Conversely, for $x \geq 0$ and any $\varepsilon > 0$, the interval $(x - \varepsilon, x + \varepsilon)$ contains a positive number, e.g. $x + \varepsilon/2 > 0$, hence meets A . Thus $x \in \overline{A}$ and $\overline{A} = [0, \infty)$.

Step 2: $A^\circ = (0, \infty)$. For $x > 0$, choose $r = x/2 > 0$, then $(x - r, x + r) \subset (0, \infty)$, so $x \in A^\circ$.

Step 3: Boundary.

$$\partial(0, \infty) = \overline{A} \setminus A^\circ = [0, \infty) \setminus (0, \infty) = \{0\}.$$

Numeric Check. At 0, every interval $(-r, r)$ contains both positive and negative numbers, so it meets A and its complement.

Solved Example. Boundary of $\mathbb{Z}$ in $\mathbb{R}$

Show that $\partial\mathbb{Z} = \mathbb{Z}$.

Solution: We prove two inclusions.

(i) $\mathbb{Z} \subseteq \partial\mathbb{Z}$. Fix $n \in \mathbb{Z}$. Let $\varepsilon > 0$. Then $(n - \varepsilon, n + \varepsilon)$ contains n itself, so it meets $\mathbb{Z}$. Also it contains a non-integer (e.g. $n + \varepsilon/2$), hence it meets $\mathbb{R} \setminus \mathbb{Z}$. Thus every neighborhood meets both sides, so $n \in \partial\mathbb{Z}$.

(ii) $\partial\mathbb{Z} \subseteq \mathbb{Z}$. Let $x \notin \mathbb{Z}$. Define the distance from x to $\mathbb{Z}$:

$$d := \text{dist}(x, \mathbb{Z}) = \inf\{|x - n| : n \in \mathbb{Z}\}.$$

In $\mathbb{R}$, the nearest integer exists; hence $d > 0$. Choose $r = \frac{d}{2} > 0$. Then $(x - r, x + r)$ contains no integer (otherwise $|x - n| < r = d/2$ contradicts minimality of d). So $(x - r, x + r) \cap \mathbb{Z} = \emptyset$, meaning $x \notin \overline{\mathbb{Z}}$ and hence $x \notin \partial\mathbb{Z}$. Therefore $\partial\mathbb{Z} \subseteq \mathbb{Z}$.

Combining (i) and (ii), $\partial\mathbb{Z} = \mathbb{Z}$.

Numeric Viewpoint: If $x = 2.3$, the nearest integer is 2 or 3, and $d = \min\{0.3, 0.7\} = 0.3$. Taking $r = 0.15$, the interval $(2.15, 2.45)$ contains no integers, so x is not a boundary point.

Solved Example. Boundary of a Dense Set: Irrationals $\mathbb{R} \setminus \mathbb{Q}$

Show that $\partial(\mathbb{R} \setminus \mathbb{Q}) = \mathbb{R}$.

Solution. Let $A = \mathbb{R} \setminus \mathbb{Q}$. A point x lies in ∂A iff every open interval around x meets both A and $\mathbb{R} \setminus A = \mathbb{Q}$. But $\mathbb{Q}$ is dense in $\mathbb{R}$ and $\mathbb{R} \setminus \mathbb{Q}$ is also dense in $\mathbb{R}$: every interval contains rationals and irrationals. Hence every $x \in \mathbb{R}$ is a boundary point, so

$$\partial(\mathbb{R} \setminus \mathbb{Q}) = \mathbb{R}.$$

Numerical Intuition: Given an interval like $(1.4142, 1.4143)$ you can find a rational by truncating decimals, and you can find an irrational by adding $\sqrt{2}/10^6$ (or any known irrational scaled small enough).

Solved Example. Boundary of a Union

Compute $\partial((0, 1) \cup (2, 3))$ in $\mathbb{R}$.

Solution: Let $A = (0, 1) \cup (2, 3)$.

Step 1: Closure and Interior. The interior is $A^\circ = A$ since A is open (union of open intervals). The closure is

$$\overline{A} = \overline{(0, 1)} \cup \overline{(2, 3)} = [0, 1] \cup [2, 3].$$

Step 2: Boundary.

$$\partial A = \overline{A} \setminus A^\circ = ([0, 1] \cup [2, 3]) \setminus ((0, 1) \cup (2, 3)) = \{0, 1, 2, 3\}.$$

Numeric Check (Neighborhood Test): Take 1. Every $(1 - \varepsilon, 1 + \varepsilon)$ hits $(0, 1)$ on the left and hits $(1, \infty) \subset \mathbb{R} \setminus A$ on the right, so 1 is boundary. But take 0.5: choose $\varepsilon = 0.2$, then $(0.3, 0.7) \subset (0, 1) \subset A$, so 0.5 is interior.

Solved Example. Boundary of the Cantor Set: ∂C

Let $C \subset [0, 1]$ be the Cantor middle-third set. Show that $\partial C = C$.

Solution: We use two classical facts:

- 1) C is closed in $\mathbb{R}$ (hence $\overline{C} = C$).
- 2) C has empty interior in $\mathbb{R}$ (i.e. $C^\circ = \emptyset$).

Assuming these, compute

$$\partial C = \overline{C} \setminus C^\circ = C \setminus \emptyset = C.$$

Why $C^\circ = \emptyset$ (sketch). Every nontrivial interval contains points removed at some stage of the construction, so no interval can sit inside C .

Numeric Viewpoint (Approximation): At stage n , C_n is a union of 2^n closed intervals of length 3^{-n} . Pick any point of C and any radius $r > 0$; take n large with $3^{-n} < r$. Then in $(x - r, x + r)$ you can find points of C_n and points of the removed middle thirds, showing every neighborhood meets both C and its complement.

Solved Example. Boundary in $\mathbb{R}^2$: Open Disk & Circle

Let $B = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < 1\}$. Show that $\partial B = \{(x, y) : x^2 + y^2 = 1\}$.

Solution: Let $S = \{(x, y) : x^2 + y^2 = 1\}$.

(i) $S \subseteq \partial B$. Take $p \in S$ and any $r > 0$. Move slightly inward and outward along the radial line:

$$p_- = (1 - \frac{r}{4})p, \quad p_+ = (1 + \frac{r}{4})p.$$

Then

$$\|p_-\| = 1 - \frac{r}{4} < 1 \Rightarrow p_- \in B, \quad \|p_+\| = 1 + \frac{r}{4} > 1 \Rightarrow p_+ \notin B.$$

Also

$$\|p_- - p\| = \frac{r}{4}\|p\| = \frac{r}{4} < r, \quad \|p_+ - p\| = \frac{r}{4}\|p\| = \frac{r}{4} < r,$$

so both p_-, p_+ lie in $B(p, r)$. Hence $B(p, r)$ meets both B and $\mathbb{R}^2 \setminus B$, so $p \in \partial B$.

(ii) $\partial B \subseteq S$. If $\|p\| < 1$ then p is interior: choose $r = 1 - \|p\| > 0$ and show $B(p, r) \subset B$ by the triangle inequality. So such p are not boundary. If $\|p\| > 1$ then p is exterior: choose $r = \|p\| - 1 > 0$ and show $B(p, r) \subset \mathbb{R}^2 \setminus \overline{B}$. Thus not boundary. Therefore boundary points must satisfy $\|p\| = 1$, i.e. $p \in S$.

Combining (i) and (ii), $\partial B = S$.

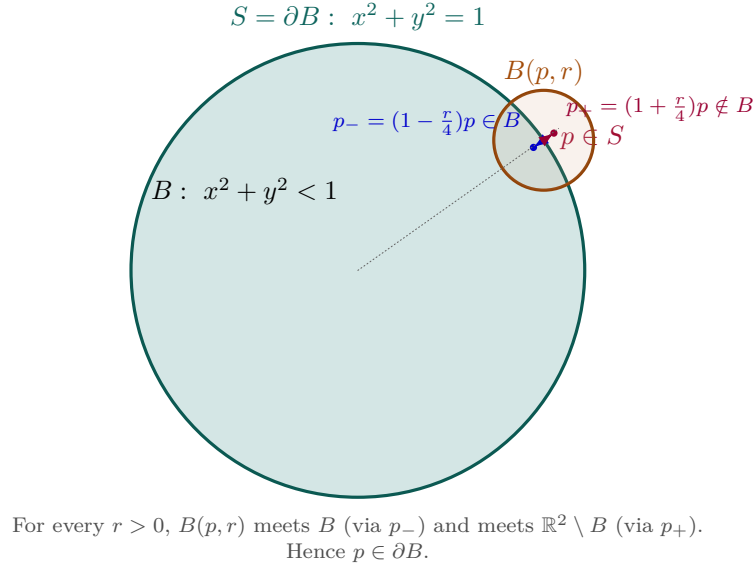


Figure 5: Boundary Test on the Unit Disk: A Neighborhood Around $p \in S$ hits both B and $\mathbb{R}^2 \setminus B$.

2 Orbits Under Iteration: Bounded, Escaping, Periodic, Chaotic

Idea / Intuition.

Given x_0 and a map $f : X \rightarrow X$, we study the orbit

$$x_0, f(x_0), f^2(x_0), \dots$$

Typical behaviors include: convergence to a fixed point, convergence to a cycle, escape to infinity (in non-compact spaces), or complicated/aperiodic motion (often called “chaotic” in dynamical systems).

2.1 Core definitions

Definition. Iterates and Orbit

Let X be a set and $f : X \rightarrow X$. Define $f^0 = \text{id}_X$ and $f^{n+1} = f \circ f^n$. For $x \in X$, the *forward orbit* is

$$\mathcal{O}^+(x) = \{x, f(x), f^2(x), \dots\}.$$

Definition. Fixed and Periodic Points

A point x is *fixed* if $f(x) = x$. A point x is *periodic* if there exists $p \in \mathbb{N}$ such that $f^p(x) = x$. The smallest such p is the *period* of x .

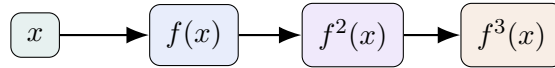
Definition. Bounded and Escaping Orbits

Let (X, d) be a metric space and fix a reference point $x_* \in X$.

- The orbit of x is *bounded* if $\{d(f^n(x), x_*) : n \in \mathbb{N}\}$ is bounded in $\mathbb{R}$.
- If $X = \mathbb{R}^m$ (or $\mathbb{C}$) with its usual norm, we say the orbit *escapes to infinity* if $\|f^n(x)\| \rightarrow \infty$.

Common Pitfall. Important Correction

A bounded orbit does *not* have to be eventually periodic. Bounded aperiodic orbits exist (e.g., irrational rotations on the circle, and many chaotic maps on intervals). Periodicity is one special bounded behavior, not the general rule.

2.2 Orbit as Repeated Application

Orbit = Repeated Application of f

2.3 Rotation Example: Cyclic vs Non-Cyclic Behavior**Definition.** Rotation on the Circle

Let $\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$ be the circle group (angles modulo 2π). For a fixed $\theta \in \mathbb{R}$ define the rotation map

$$R_\theta : \mathbb{T} \rightarrow \mathbb{T}, \quad R_\theta([\varphi]) = [\varphi + \theta],$$

where $[\varphi]$ denotes the equivalence class of φ modulo $2\pi\mathbb{Z}$. Equivalently, after choosing representatives in $[0, 2\pi)$, this is “add θ and reduce modulo 2π .”

Solved Example. Rotation by 90° Has Minimal Period 4

Let $\theta = \frac{\pi}{2}$. Show that every point of $\mathbb{T}$ is periodic and has *minimal* period 4.

Solution. For any $[\varphi] \in \mathbb{T}$,

$$R_{\pi/2}^4([\varphi]) = [\varphi + 4 \cdot \frac{\pi}{2}] = [\varphi + 2\pi] = [\varphi].$$

Hence every point is periodic with period dividing 4.

To see the minimal period is exactly 4, note that for $k = 1, 2, 3$ we have

$$k \cdot \frac{\pi}{2} \notin 2\pi\mathbb{Z}.$$

Therefore

$$R_{\pi/2}^k([\varphi]) = [\varphi + k\frac{\pi}{2}] \neq [\varphi] \quad (k = 1, 2, 3),$$

for every $[\varphi] \in \mathbb{T}$. Thus the minimal period is 4, and the orbit is

$$[\varphi], [\varphi + \frac{\pi}{2}], [\varphi + \pi], [\varphi + \frac{3\pi}{2}], [\varphi].$$

Solved Example. Irrational Rotation Has no Periodic Points

Assume $\theta/2\pi \notin \mathbb{Q}$. Show that R_θ has no periodic points.

Solution. Suppose $[\varphi] \in \mathbb{T}$ is periodic. Then there exists $p \in \mathbb{N}$ such that

$$R_\theta^p([\varphi]) = [\varphi].$$

But

$$R_\theta^p([\varphi]) = [\varphi + p\theta],$$

so $[\varphi + p\theta] = [\varphi]$, which is equivalent to

$$p\theta \in 2\pi\mathbb{Z}.$$

Hence $p\theta = 2\pi k$ for some $k \in \mathbb{Z}$, so

$$\frac{\theta}{2\pi} = \frac{k}{p} \in \mathbb{Q},$$

contradicting the assumption that $\theta/2\pi$ is irrational. Therefore R_θ has no periodic points.

Solved Example. Compute Iterates Modulo 2π

Let $\theta = \frac{\pi}{2}$ and $\varphi_0 = \frac{\pi}{6}$. Compute

$$\varphi_n \equiv \varphi_0 + n\theta \pmod{2\pi} \quad \text{for } n = 0, 1, 2, 3, 4, 5,$$

and list the orbit in $[0, 2\pi)$.

Solution. We compute and reduce to $[0, 2\pi)$:

$$\begin{aligned} \varphi_0 &= \frac{\pi}{6}, \\ \varphi_1 &= \frac{\pi}{6} + \frac{\pi}{2} = \frac{4\pi}{6} = \frac{2\pi}{3}, \\ \varphi_2 &= \frac{\pi}{6} + \pi = \frac{7\pi}{6}, \\ \varphi_3 &= \frac{\pi}{6} + \frac{3\pi}{2} = \frac{10\pi}{6} = \frac{5\pi}{3}, \\ \varphi_4 &= \frac{\pi}{6} + 2\pi = \frac{13\pi}{6} \equiv \frac{\pi}{6} \pmod{2\pi}, \\ \varphi_5 &= \varphi_4 + \frac{\pi}{2} = \frac{2\pi}{3}. \end{aligned}$$

Hence the orbit (in $[0, 2\pi)$) is

$$\frac{\pi}{6} \mapsto \frac{2\pi}{3} \mapsto \frac{7\pi}{6} \mapsto \frac{5\pi}{3} \mapsto \frac{\pi}{6} \mapsto \dots$$

and it has minimal period 4.

If we also list points on the unit circle $z_n = (\cos \varphi_n, \sin \varphi_n)$, then

$$\begin{aligned} z_0 &= \left(\cos \frac{\pi}{6}, \sin \frac{\pi}{6} \right) = \left(\frac{\sqrt{3}}{2}, \frac{1}{2} \right), \\ z_1 &= \left(\cos \frac{2\pi}{3}, \sin \frac{2\pi}{3} \right) = \left(-\frac{1}{2}, \frac{\sqrt{3}}{2} \right), \\ z_2 &= \left(\cos \frac{7\pi}{6}, \sin \frac{7\pi}{6} \right) = \left(-\frac{\sqrt{3}}{2}, -\frac{1}{2} \right), \\ z_3 &= \left(\cos \frac{5\pi}{3}, \sin \frac{5\pi}{3} \right) = \left(\frac{1}{2}, -\frac{\sqrt{3}}{2} \right). \end{aligned}$$

Solved Example. Rotation by 120° : Find the Period Numerically

Take $\theta = \frac{2\pi}{3}$ and $\varphi_0 = 1$ (radians). Compute $\varphi_1, \dots, \varphi_7$ where

$$\varphi_{n+1} \equiv \varphi_n + \theta \pmod{2\pi}.$$

Determine the smallest $p \geq 1$ such that $\varphi_p \equiv \varphi_0 \pmod{2\pi}$.

Solution. We compute iterates (write $\theta = \frac{2\pi}{3}$):

$$\begin{aligned} \varphi_1 &\equiv 1 + \frac{2\pi}{3}, \\ \varphi_2 &\equiv 1 + \frac{4\pi}{3}, \\ \varphi_3 &\equiv 1 + 2\pi \equiv 1, \\ \varphi_4 &\equiv 1 + \frac{2\pi}{3}, \quad \varphi_5 \equiv 1 + \frac{4\pi}{3}, \quad \varphi_6 \equiv 1, \quad \varphi_7 \equiv 1 + \frac{2\pi}{3}. \end{aligned}$$

Thus the orbit repeats after $p = 3$, and the minimal period is 3.

Solved Example. Negative Rotation and Reduction

Let $\theta = -\frac{\pi}{3}$ and $\varphi_0 = \frac{5\pi}{4}$. Compute $\varphi_n \in [0, 2\pi)$ for $n = 1, \dots, 6$, where

$$\varphi_{n+1} \equiv \varphi_n + \theta \pmod{2\pi}.$$

Solution. Compute step by step and keep values in $[0, 2\pi)$:

$$\begin{aligned} \varphi_1 &= \frac{5\pi}{4} - \frac{\pi}{3} = \frac{15\pi - 4\pi}{12} = \frac{11\pi}{12}, \\ \varphi_2 &= \frac{11\pi}{12} - \frac{\pi}{3} = \frac{11\pi - 4\pi}{12} = \frac{7\pi}{12}, \\ \varphi_3 &= \frac{7\pi}{12} - \frac{\pi}{3} = \frac{3\pi}{12} = \frac{\pi}{4}, \\ \varphi_4 &= \frac{\pi}{4} - \frac{\pi}{3} = -\frac{\pi}{12} \equiv 2\pi - \frac{\pi}{12} = \frac{23\pi}{12}, \\ \varphi_5 &= \frac{23\pi}{12} - \frac{\pi}{3} = \frac{19\pi}{12}, \\ \varphi_6 &= \frac{19\pi}{12} - \frac{\pi}{3} = \frac{15\pi}{12} = \frac{5\pi}{4}. \end{aligned}$$

Hence $\varphi_6 = \varphi_0$, so the orbit is periodic with minimal period 6 (indeed $\theta/2\pi = -1/6$).

Solved Example. Minimal Period from a Rational Angle

Suppose $\theta = 2\pi \cdot \frac{7}{20}$.

1. Find the minimal $p \geq 1$ such that $R_\theta^p = \text{id}_{\mathbb{T}}$.
2. Verify numerically that $p\theta \in 2\pi\mathbb{Z}$.

Solution. We need the smallest p such that $p\theta \in 2\pi\mathbb{Z}$:

$$p\theta = p \cdot 2\pi \frac{7}{20} = 2\pi \cdot \frac{7p}{20} \in 2\pi\mathbb{Z} \iff \frac{7p}{20} \in \mathbb{Z}.$$

Since $\gcd(7, 20) = 1$, this happens first when $20 \mid p$, i.e. $p = 20$. Numerical verification:

$$20\theta = 20 \cdot 2\pi \frac{7}{20} = 14\pi = 2\pi \cdot 7 \in 2\pi\mathbb{Z}.$$

Solved Example. Detect θ and the Period from Orbit Data

An orbit (reduced to $[0, 2\pi)$) begins

$$\varphi_0 = 0.2, \quad \varphi_1 = 2.2944, \quad \varphi_2 = 4.3888, \quad \varphi_3 = 0.2, \dots$$

1. Estimate θ numerically.
2. Determine the minimal period of φ_0 under R_θ .

Solution.

1. Since $\varphi_1 \equiv \varphi_0 + \theta \pmod{2\pi}$, we estimate

$$\theta \equiv \varphi_1 - \varphi_0 = 2.2944 - 0.2 = 2.0944 \text{ (radians)}.$$

Check with the next step:

$$\varphi_2 - \varphi_1 = 4.3888 - 2.2944 = 2.0944,$$

so the increment is consistent; thus $\theta \approx 2.0944$.

2. We have $\varphi_3 = \varphi_0$, so the period divides 3. Since $\varphi_1 \neq \varphi_0$ and $\varphi_2 \neq \varphi_0$, the minimal period is $p = 3$. (Indeed $2.0944 \approx \frac{2\pi}{3} \approx 2.094395\dots$)

Solved Example. Approximate Return for an “Irrational” Rotation

Let $\theta = 2\pi(\sqrt{2} - 1)$ and $\varphi_0 = 0$. Define

$$\varphi_n \equiv n\theta \pmod{2\pi}, \quad n \geq 1.$$

Compute φ_n for $n = 1, \dots, 50$ and find the smallest n such that φ_n is within 0.05 radians of 0 in circular distance, i.e.

$$\min(\varphi_n, 2\pi - \varphi_n) < 0.05.$$

Solution. Write $\alpha = \frac{\theta}{2\pi} = \sqrt{2} - 1$ (irrational). Then

$$\varphi_n \equiv 2\pi\{n\alpha\} \quad \text{where } \{x\} \text{ is the fractional part.}$$

Being within 0.05 radians of 0 means

$$2\pi\{n\alpha\} < 0.05 \quad \text{or} \quad 2\pi(1 - \{n\alpha\}) < 0.05.$$

Equivalently,

$$\{n\alpha\} < \frac{0.05}{2\pi} \approx 0.00796 \quad \text{or} \quad 1 - \{n\alpha\} < 0.00796.$$

A quick computation for $n \leq 50$ (e.g. by calculator / Python) shows the smallest such n is

$$n = 29,$$

with

$$\varphi_{29} \approx 0.0494 \text{ radians,}$$

so $\min(\varphi_{29}, 2\pi - \varphi_{29}) \approx 0.0494 < 0.05$.

Solved Example. Closest Approach to the Starting Point

With $\theta = 2\pi(\sqrt{2} - 1)$ and $\varphi_0 = 0$, define

$$d_n = \min(|\varphi_n - \varphi_0|, 2\pi - |\varphi_n - \varphi_0|), \quad \varphi_n \equiv n\theta \pmod{2\pi}.$$

Compute d_n for $n = 1, \dots, 200$, and determine the n for which d_n is minimal, together with that minimum value.

Solution. Since $\varphi_0 = 0$ and $\varphi_n \in [0, 2\pi)$, this simplifies to

$$d_n = \min(\varphi_n, 2\pi - \varphi_n).$$

A direct computation for $1 \leq n \leq 200$ gives the closest return at

$$n = 70, \quad d_{70} \approx 0.0210 \text{ radians.}$$

(Any equivalent numeric answer obtained by correct computation is acceptable; the purpose is to practice “closest return” under an irrational rotation.)

Solved Example. Rotation Preserves Circular Distance: Numeric Check

Let $\varphi = 0.2$, $\psi = 5.9$, and $\theta = 1.1$. Define

$$d_{\mathbb{T}}([\varphi], [\psi]) = \min(|\varphi - \psi|, 2\pi - |\varphi - \psi|).$$

1. Compute $d_{\mathbb{T}}([\varphi], [\psi])$.
2. Compute $d_{\mathbb{T}}(R_{\theta}([\varphi]), R_{\theta}([\psi]))$.
3. Verify numerically that these two values are equal.

Solution. Compute $|\varphi - \psi| = |0.2 - 5.9| = 5.7$. Also $2\pi \approx 6.283185$, so

$$2\pi - |\varphi - \psi| \approx 6.283185 - 5.7 = 0.583185.$$

Hence

$$d_{\mathbb{T}}([\varphi], [\psi]) = \min(5.7, 0.583185) = 0.583185 \text{ (approx).}$$

Now rotate:

$$R_{\theta}([\varphi]) = [\varphi + \theta] = [1.3], \quad R_{\theta}([\psi]) = [7.0] \equiv [7.0 - 2\pi] \approx [0.716815].$$

Thus the distance becomes

$$|1.3 - 0.716815| = 0.583185,$$

and

$$2\pi - 0.583185 \approx 5.7,$$

so

$$d_{\mathbb{T}}(R_{\theta}([\varphi]), R_{\theta}([\psi])) = 0.583185 = d_{\mathbb{T}}([\varphi], [\psi]).$$

This confirms numerically that rotation preserves circular distance.

Solved Example. Reverse Engineering θ from a Given Period

Assume a point has *minimal* period $p = 12$ under R_{θ} , with $\theta \in (0, 2\pi)$. List all θ of the form $\theta = 2\pi \cdot \frac{k}{12}$ that yield minimal period 12.

Solution. For $\theta = 2\pi \frac{k}{12}$, the order (minimal period) is

$$p = \frac{12}{\gcd(12, k)}.$$

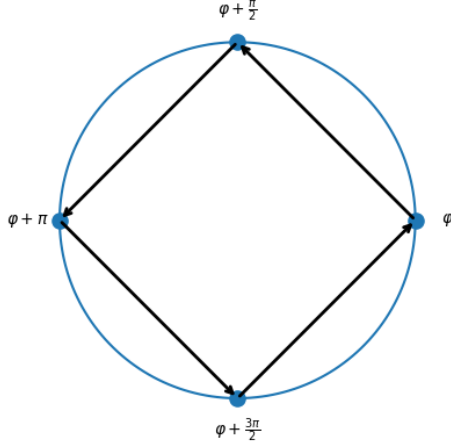
We want $p = 12$, i.e. $\gcd(12, k) = 1$. With $1 \leq k \leq 11$, the integers coprime to 12 are

$$k \in \{1, 5, 7, 11\}.$$

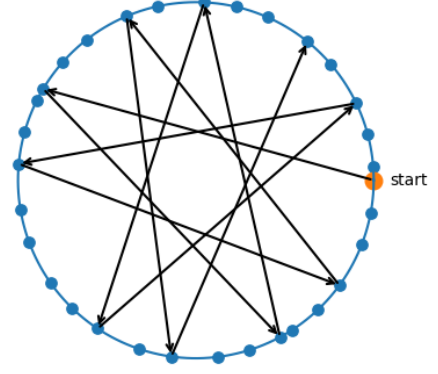
Hence the angles are

$$\theta \in \left\{ \frac{2\pi}{12}, \frac{10\pi}{12}, \frac{14\pi}{12}, \frac{22\pi}{12} \right\} = \left\{ \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6} \right\}.$$

Rotation on $\mathbb{T}$: $\theta = \pi/2$ (4-cycle)



Rotation on $\mathbb{T}$: $\theta/2\pi \notin \mathbb{Q}$ (no periodic points)



2.4 Explanation of the Diagram (Rotation on $\mathbb{T}$).

The figure has two panels, each representing the circle

$$\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z}),$$

viewed as the unit circle in the plane.

Left panel (rational rotation $\theta = \pi/2$). Starting from a point $[\varphi] \in \mathbb{T}$ (marked on the circle), we apply the rotation

$$R_{\pi/2}([\varphi]) = [\varphi + \frac{\pi}{2}].$$

The arrows show successive iterates:

$$[\varphi] \mapsto [\varphi + \frac{\pi}{2}] \mapsto [\varphi + \pi] \mapsto [\varphi + \frac{3\pi}{2}] \mapsto [\varphi].$$

After four steps we return exactly to the starting point, i.e. $R_{\pi/2}^4 = \text{id}$ on $\mathbb{T}$, and the four marked points form a 4-cycle (a square inscribed in the circle). This visually encodes that every point has minimal period 4 under $R_{\pi/2}$.

Right panel (irrational rotation $\theta/2\pi \notin \mathbb{Q}$). The orange point marks the starting position. The blue dots represent successive iterates

$$[\varphi], [\varphi + \theta], [\varphi + 2\theta], \dots,$$

and the arrows indicate the first few transitions between consecutive iterates. Because $\theta/2\pi$ is irrational, no iterate returns exactly to the start: there is no $p \geq 1$ with $R_{\theta}^p([\varphi]) = [\varphi]$. Hence the orbit never forms a finite cycle; instead it keeps producing new points around the circle (and, in fact, such orbits are dense in $\mathbb{T}$).

2.5 Quadratic Map on $\mathbb{C}$: Bounded vs Escaping Orbits

Definition. Quadratic Family

Fix $c \in \mathbb{C}$ and define $f_c : \mathbb{C} \rightarrow \mathbb{C}$ by

$$f_c(z) = z^2 + c.$$

Given a starting point $z_0 \in \mathbb{C}$, its (forward) orbit under f_c is the sequence

$$z_{n+1} = f_c(z_n) = z_n^2 + c, \quad n = 0, 1, 2, \dots$$

Idea / Intuition.

A fundamental dichotomy for this family is whether an orbit stays *bounded* or *escapes to infinity*. A standard (and very useful) fact is the following *escape criterion*:

If $|z_n| > 2$ for some n , then $|z_k| \rightarrow \infty$ as $k \rightarrow \infty$.

Indeed, if $|z| > 2$ and $|c| \leq 2$, then

$$|f_c(z)| = |z^2 + c| \geq |z|^2 - |c| \geq |z|^2 - 2 > |z|.$$

So once the orbit enters the region outside the circle $|z| = 2$, its modulus increases rapidly and the orbit must diverge. Therefore, to check boundedness, it is enough to iterate and watch whether the orbit ever leaves the disk $|z| \leq 2$.

Lemma. Escape Radius Lemma

Let $f_c(z) = z^2 + c$. If $|z_n| > R$ for some n , where

$$R := \max\{2, |c| + 1\},$$

then $|z_{n+k}| \rightarrow \infty$ as $k \rightarrow \infty$. In particular, the orbit escapes to infinity.

Proof (full, inequality-driven). Assume $|z_n| > R \geq |c| + 1$. Then

$$|z_{n+1}| = |z_n^2 + c| \geq |z_n|^2 - |c| > |z_n|^2 - (|z_n| - 1) = |z_n|^2 - |z_n| + 1.$$

Since $|z_n| > 1$, we have $|z_n|^2 - |z_n| + 1 > |z_n|$. Hence $|z_{n+1}| > |z_n|$. Repeating this, the sequence $|z_{n+k}|$ is strictly increasing and, moreover, eventually satisfies $|z_{n+k+1}| \geq |z_{n+k}|^2 - |c|$, which forces divergence to $+\infty$ because squaring dominates subtraction of a constant. Therefore the orbit escapes. $\square$

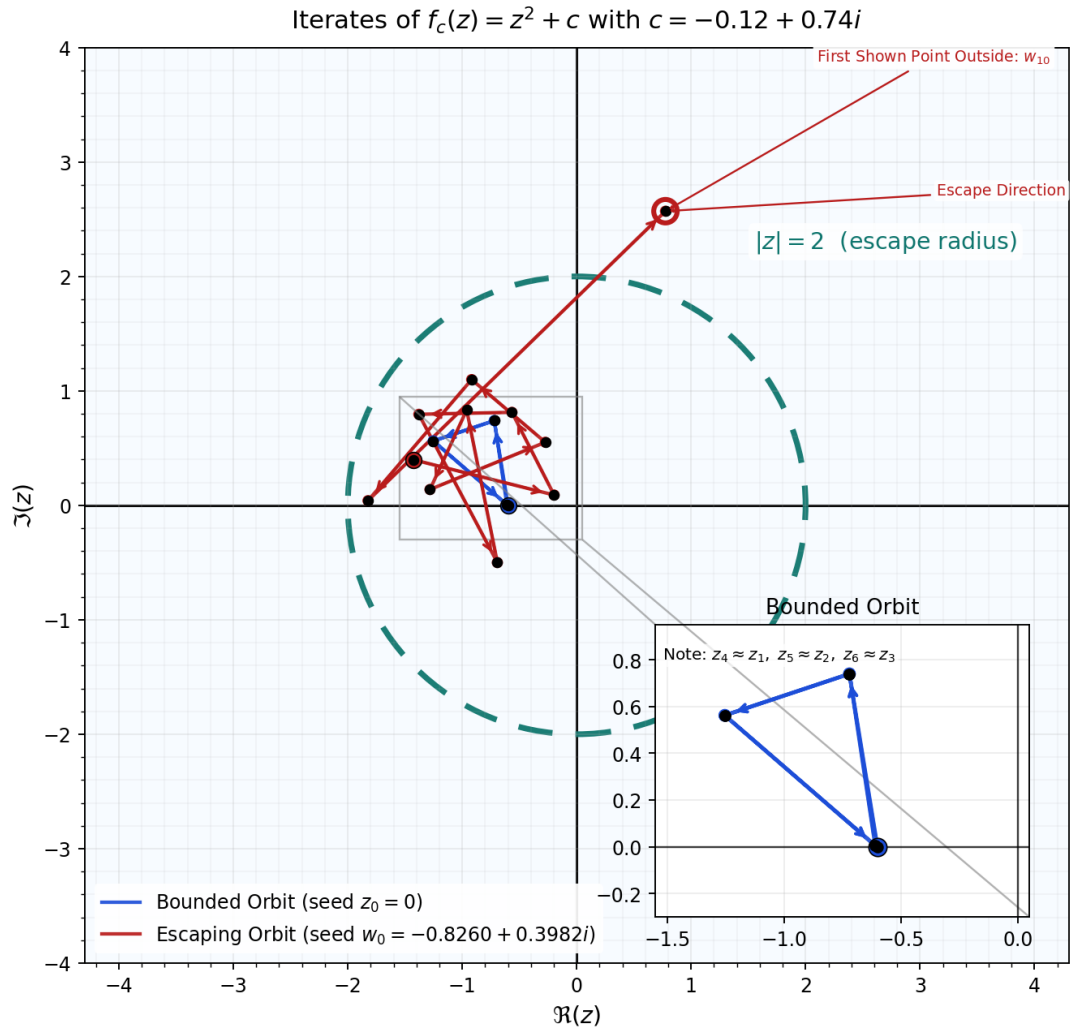


Figure 6: Actual iterates of $f_c(z) = z^2 + c$ with $c = -0.12 + 0.74i$. The blue orbit (seed $z_0 = 0$) stays bounded inside the escape disk, while the red orbit (seed $w_0 = -0.8260 + 0.3982i$) eventually crosses $|z| = 2$ at w_{10} , after which the modulus grows rapidly and the orbit diverges.

How to Read the Diagram (Mathematical Meaning): The picture represents the dynamics of the quadratic map

$$f_c(z) = z^2 + c, \quad c = -0.12 + 0.74i,$$

acting on the complex plane $\mathbb{C}$. Starting from a seed $z_0 \in \mathbb{C}$, we generate the *orbit*

$$z_{n+1} = f_c(z_n) = z_n^2 + c, \quad n = 0, 1, 2, \dots$$

Each plotted dot is an iterate z_n , and the line segments connect consecutive iterates.

Axes and Geometry: The horizontal axis is $\Re(z)$ and the vertical axis is $\Im(z)$. Thus each point in the plot is a complex number $z = x + iy$ drawn at (x, y) .

The Dashed Circle $|z| = 2$ (Escape Radius): The dashed circle marks the set $\{z \in \mathbb{C} : |z| = 2\}$. It is important because of the standard *escape criterion* for the quadratic family: once an orbit satisfies $|z_n| > 2$ at some step n , the orbit will diverge to infinity (i.e. $|z_k| \rightarrow \infty$ as $k \rightarrow \infty$). Intuitively, outside this circle the squaring term dominates and makes the modulus grow rapidly. Hence, in numerical experiments, monitoring whether an iterate leaves $|z| \leq 2$ is a reliable test for escaping behavior.

Bounded Orbit (Inside the Escape Disk): One plotted orbit starts at the critical point $z_0 = 0$ and continues as

$$z_1 = c, \quad z_2 = c^2 + c, \quad z_3 = (c^2 + c)^2 + c, \quad \dots$$

In the diagram these points remain within the disk $|z| \leq 2$ for the displayed iterates; geometrically, the orbit moves around inside the dashed circle without crossing it. This indicates *bounded* behavior for this initial seed (at least for the observed iterates), and in complex dynamics this boundedness of the critical orbit is the key feature that distinguishes connected vs. disconnected Julia sets and is central in the study of the Mandelbrot set.

Escaping Orbit (Eventual Exit From $|z| = 2$): A second orbit begins at a different seed $w_0 = -0.8260 + 0.3982i$. The iterates $w_1, w_2, \dots$ initially stay near the central region, but eventually one iterate satisfies $|w_{10}| > 2$, which is explicitly marked in the figure. From this point onward, subsequent iterates move quickly outward; the arrow indicates the direction of further growth. By the escape criterion, crossing $|z| = 2$ implies the orbit escapes $|w_n| \rightarrow \infty$ ($n \rightarrow \infty$).

Key Insight (Bounded vs. Escaping): The main conceptual message of the diagram is the dichotomy: either the orbit stays bounded (never leaving $|z| \leq 2$), or once it leaves $|z| = 2$ it escapes to infinity. Thus the complex plane is dynamically partitioned into initial conditions with bounded orbits (forming the *filled Julia set* K_c) and initial conditions whose orbits escape. The boundary between these behaviors is the *Julia set* $J_c = \partial K_c$.

Definition. Filled Julia Set (Bounded Orbit Set)

The *filled Julia set* of f_c is

$$K_c := \{z \in \mathbb{C} : \{f_c^n(z)\}_{n \geq 0} \text{ is bounded}\}.$$

Its boundary $J_c := \partial K_c$ is the *Julia set*.

Solved Example. Periodic Orbit for c

Let $f(z) = z^2 - 1$ (so $c = -1$) and start at $z_0 = 0$.

1. Compute z_1, z_2, z_3, z_4 explicitly.
2. Prove that the orbit becomes periodic and find its *minimal period*.
3. Decide whether the orbit is bounded or escaping.

Solution. We iterate the recurrence $z_{n+1} = z_n^2 - 1$.

Step 1: compute the first few iterates.

$$z_0 = 0, \quad z_1 = z_0^2 - 1 = 0^2 - 1 = -1.$$

Next,

$$z_2 = z_1^2 - 1 = (-1)^2 - 1 = 1 - 1 = 0.$$

Continuing,

$$z_3 = z_2^2 - 1 = 0^2 - 1 = -1, \quad z_4 = z_3^2 - 1 = (-1)^2 - 1 = 0.$$

So we see the pattern

$$z_0 = 0, \quad z_1 = -1, \quad z_2 = 0, \quad z_3 = -1, \quad z_4 = 0, \dots$$

Step 2: prove periodicity and minimal period. From the computations we already have

$$z_2 = 0 = z_0.$$

Because the recurrence is deterministic, if a value repeats then the entire future repeats. More formally: if $z_m = z_n$, then $z_{m+1} = f(z_m) = f(z_n) = z_{n+1}$, and by induction $z_{m+k} = z_{n+k}$ for all $k \geq 0$. Applying this with $m = 2$ and $n = 0$ gives

$$z_{2+k} = z_k \quad \text{for all } k \geq 0,$$

so the orbit is periodic with period dividing 2.

To find the *minimal* period, check whether period 1 is possible: a period-1 orbit would require $z_1 = z_0$, but here $z_1 = -1 \neq 0$. Hence the minimal period is exactly 2.

Step 3: bounded vs escaping. A periodic orbit takes only finitely many values, hence it is bounded. Concretely,

$$\{z_n : n \geq 0\} \subseteq \{0, -1\} \quad \Rightarrow \quad |z_n| \leq 1 \text{ for all } n.$$

Therefore the orbit is bounded (and in fact is a 2-cycle $0 \leftrightarrow -1$).

Solved Example. Escaping Orbit for c

Let $f(z) = z^2$ (so $c = 0$) and start at $z_0 = 2$.

1. Compute z_1, z_2, z_3, z_4 .
2. Find a closed-form expression for z_n .
3. Prove that $|z_n| \rightarrow \infty$ as $n \rightarrow \infty$.

Solution. Here $z_{n+1} = z_n^2$ with $z_0 = 2$.

Step 1: compute initial iterates.

$$z_1 = 2^2 = 4, \quad z_2 = 4^2 = 16, \quad z_3 = 16^2 = 256, \quad z_4 = 256^2 = 65536.$$

The numbers grow extremely quickly.

Step 2: closed form. We claim

$$z_n = 2^{2^n} \quad (n \geq 0).$$

Proof by induction. For $n = 0$, $z_0 = 2 = 2^{2^0}$. Assume $z_n = 2^{2^n}$. Then

$$z_{n+1} = z_n^2 = (2^{2^n})^2 = 2^{2 \cdot 2^n} = 2^{2^{n+1}}.$$

So the formula holds for all n .

Step 3: escape to infinity. Since $2^{2^n} \rightarrow \infty$ as $n \rightarrow \infty$, we have $|z_n| \rightarrow \infty$. Hence the orbit escapes.

Solved Example. Numerical Boundedness Test for c

Let $f_c(z) = z^2 + c$ with $c = 0.3 + 0.5i$ and start at $z_0 = 0$.

1. Compute z_1 and z_2 exactly in the form $a + bi$.
2. Compute $|z_1|$ and $|z_2|$ (numerically to 3 decimal places).
3. State a concrete algorithm (an iteration test) to decide whether the orbit escapes.
4. Explain why crossing the radius 2 guarantees escape.

Solution.

Step 1: compute the first iterates. Since $z_0 = 0$,

$$z_1 = z_0^2 + c = c = 0.3 + 0.5i.$$

Next,

$$z_2 = z_1^2 + c = (0.3 + 0.5i)^2 + (0.3 + 0.5i).$$

Compute the square:

$$(0.3 + 0.5i)^2 = 0.3^2 + 2(0.3)(0.5)i + (0.5i)^2 = 0.09 + 0.30i - 0.25 = -0.16 + 0.30i.$$

Hence

$$z_2 = (-0.16 + 0.30i) + (0.3 + 0.5i) = 0.14 + 0.80i.$$

Step 2: compute moduli.

$$|z_1| = \sqrt{0.3^2 + 0.5^2} = \sqrt{0.09 + 0.25} = \sqrt{0.34} \approx 0.583.$$

$$|z_2| = \sqrt{0.14^2 + 0.80^2} = \sqrt{0.0196 + 0.64} = \sqrt{0.6596} \approx 0.812.$$

So the orbit has not escaped in the first two steps.

Step 3: a concrete numerical test (algorithm). Choose a maximum number of iterations N (e.g. $N = 1000$). Iterate

$$z_{n+1} = z_n^2 + c$$

and at each step compute $|z_n|$.

- If for some $n \leq N$ we find $|z_n| > 2$, then declare: *the orbit escapes*.
- If no iterate exceeds 2 up to time N , then declare: *no escape detected up to time N* (the orbit may be bounded, or it may escape after a longer time).

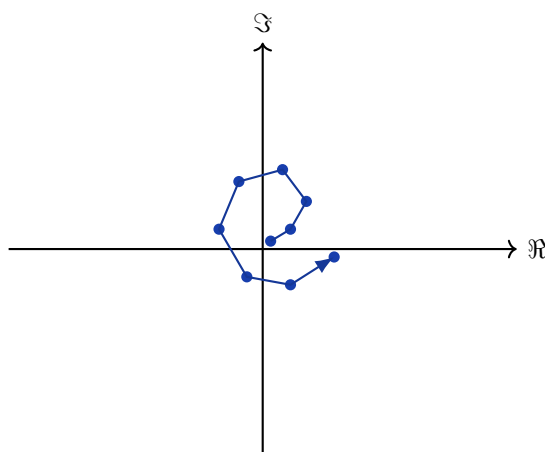
Step 4: why the threshold 2 works (escape lemma). For the quadratic family, one shows that once $|z_n| > 2$, the squaring dominates and the modulus increases. A common estimate (sufficient for computation) is: if $|z| > 2$ and $|c| \leq 2$, then

$$|f_c(z)| = |z^2 + c| \geq |z|^2 - |c| \geq |z|^2 - 2 > |z|.$$

Thus the moduli form a strictly increasing sequence after crossing 2, forcing $|z_n| \rightarrow \infty$. Since $|c| = \sqrt{0.3^2 + 0.5^2} \approx 0.583 < 2$ here, the escape lemma applies. $\square$

Remark (interpretation). For a fixed c , the set of all seeds whose orbits remain bounded is the *filled Julia set* K_c . Its boundary $J_c = \partial K_c$ is the *Julia set*. Orbits starting near J_c can behave in a visually complicated way and may take many iterations before either escaping or showing a clear bounded pattern.

2.6 Orbit Points in the Complex Plane



3 Regions in $\mathbb{R}^n$: Open, Connected, and Without Boundary

Idea / Intuition.

Informally, a “region” is an open area where you can move a little without leaving the set, and where the set is in one piece (no separated components). The standard mathematical definition in analysis is: **a region is a nonempty open connected subset of $\mathbb{R}^n$.**

3.1 Open Set

Definition. Open set in $\mathbb{R}^n$

A set $U \subseteq \mathbb{R}^n$ is open if for every $x \in U$ there exists $r > 0$ such that

$$B(x, r) = \{y \in \mathbb{R}^n : \|y - x\| < r\} \subseteq U.$$

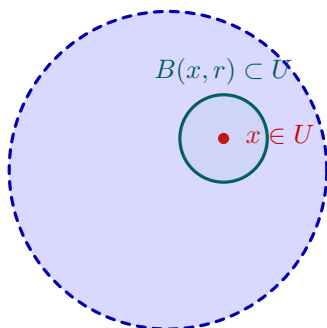
Throughout, we view $\mathbb{R}^2$ with its standard Euclidean metric

$$d(x, y) = \|x - y\|_2, \quad x, y \in \mathbb{R}^2,$$

and the open ball (disk)

$$B(x, r) = \{y \in \mathbb{R}^2 : \|y - x\| < r\}.$$

$$U = \{x^2 + y^2 < 4\} \quad \partial U : x^2 + y^2 = 4$$



Every point $x \in U$ has a small ball contained in U .
Boundary is *not* included (dashed circle).

Figure 7: Open set: the ball criterion for an open disk U .

Visualization: The picture shows an *open disk*

$$U = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < 4\}.$$

Its boundary circle $\{x^2 + y^2 = 4\}$ is drawn as a dashed curve to emphasize that boundary points are *not* included in U .

Drawn Inner Disk Meaning: A point $x \in U$ is chosen and a smaller disk $B(x, r)$ is drawn fully inside U . This is the geometric meaning of openness: every point of U is surrounded by a small neighborhood that still lies in U .

Remark 3.1. Openness is a *local* property: It is checked point-by-point via existence of a radius $r(x)$. The boundary not being included is a consequence, not the definition.

3.2 Connected & Disconnected set (two components)

Definition. Connectedness

A set $E \subseteq \mathbb{R}^2$ is *disconnected* if there exist nonempty disjoint sets U, V such that

$$E = U \cup V$$

and U, V are open in the *subspace topology* on E (equivalently, $U = E \cap G$ and $V = E \cap H$ for some open sets $G, H \subset \mathbb{R}^2$). If no such separation exists, E is *connected*.

Visualization: The diagram shows two disjoint filled disks, e.g.

$$E = D_1 \cup D_2, \quad D_1 = \overline{B}((-1.2, 0), 1), \quad D_2 = \overline{B}((1.2, 0), 1).$$

They are separated by a gap. Intuitively, E has two pieces (two connected components).

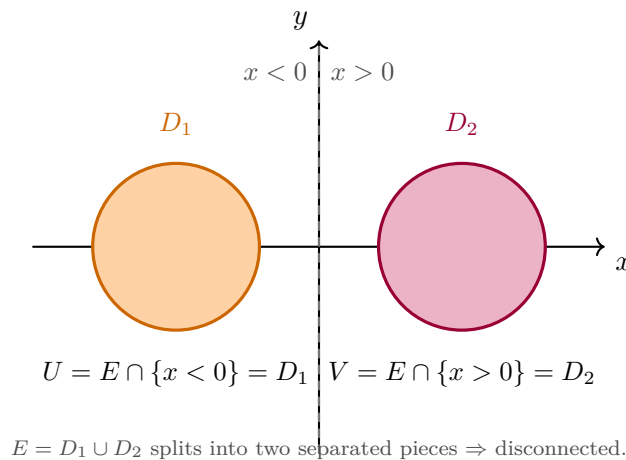


Figure 8: Disconnected set: a separation into two components.

A Clean Separation: Let

$$G = \{(x, y) : x < 0\}, \quad H = \{(x, y) : x > 0\}.$$

Then G, H are open in $\mathbb{R}^2$ and

$$U := E \cap G = D_1, \quad V := E \cap H = D_2,$$

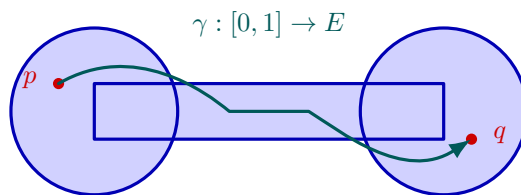
hence $E = U \cup V$ is a separation: E is disconnected.

3.3 Path-Connected Set

Definition. Path-connectedness

A set $E \subseteq \mathbb{R}^n$ is *path-connected* if for every $x, y \in E$ there exists a continuous map $\gamma : [0, 1] \rightarrow E$ with $\gamma(0) = x$ and $\gamma(1) = y$.

Path-Connected Set



Any two points can be joined by a continuous curve staying inside E . Hence E is path-connected.

Figure 9: Path-Connected Set: A Dumbbell and a Path Joining p to q .

Visualization: The dumbbell shape consists of two disks joined by a corridor. Formally one may take

$$E = \overline{B}((-a, 0), 1) \cup \left([-a, a] \times [-\varepsilon, \varepsilon]\right) \cup \overline{B}((a, 0), 1),$$

with $a > 1$ and $\varepsilon > 0$. The corridor ensures there is no “gap” between the two sides.

Why Path-Connected? Given any $p, q \in E$, one constructs a continuous curve inside E by:

- Connecting p to some point in the corridor inside its disk (straight segment stays inside the disk),
- Moving through the corridor (straight segment stays inside the corridor rectangle),
- Connecting from the corridor to q inside the other disk.

Concatenating these segments gives a continuous γ in E joining p to q .

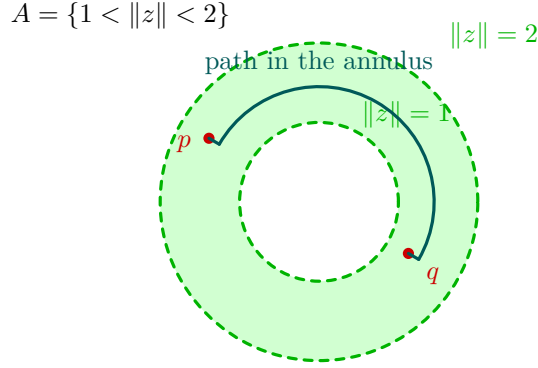
Proposition 3.2. *Every path-connected subset of $\mathbb{R}^2$ is connected.*

Idea: If E were separated into $U \cup V$, a path from a point in U to a point in V would force the continuous image $\gamma([0, 1])$ to cross both U and V , contradicting the separation by relatively open sets. $\square$

3.4 Region (Open + Connected) Via an Open Annulus

Definition. Region

A *region* in $\mathbb{R}^n$ is a nonempty open connected subset of $\mathbb{R}^n$. (Some authors require path-connected; in $\mathbb{R}^n$ open connected $\Rightarrow$ path-connected, so both match.)



A is open (dashed boundaries excluded) and connected (indeed path-connected) $\Rightarrow$ a region.

Figure 10: Region: An Open Annulus is Open and Connected (Hence path-connected).

Visualization: The open annulus

$$A = \{(x, y) \in \mathbb{R}^2 : 1 < \sqrt{x^2 + y^2} < 2\} = \{(x, y) : 1 < \|(x, y)\| < 2\}$$

is shown. Both boundary circles $\|z\| = 1$ and $\|z\| = 2$ are excluded (dashed), hence A is open.

Why It is Connected (Indeed Path-Connected)? Take any $p, q \in A$. Write them in polar coordinates:

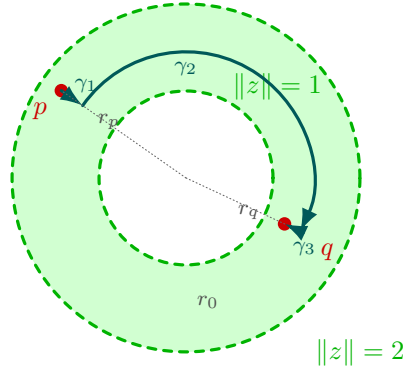
$$p = (r_p \cos \theta_p, r_p \sin \theta_p), \quad q = (r_q \cos \theta_q, r_q \sin \theta_q),$$

with $1 < r_p, r_q < 2$. Choose a radius $r_0 \in (1, 2)$ (e.g. $r_0 = \frac{r_p + r_q}{2}$). Then the path

$$\gamma(t) = \begin{cases} ((1-2t)r_p + 2tr_0)(\cos \theta_p, \sin \theta_p), & 0 \leq t \leq \frac{1}{2}, \\ r_0(\cos((2-2t)\theta_p + (2t-1)\theta_q), \sin((2-2t)\theta_p + (2t-1)\theta_q)), & \frac{1}{2} \leq t \leq 1, \end{cases}$$

stays in A and connects p to q . Hence A is path-connected, thus connected, and since it is open, it is a region.

$$A = \{(x, y) : 1 < \sqrt{x^2 + y^2} < 2\}$$



Path inside A : radial to r_0 , rotate along the circle $r = r_0$, then radial to q .
 Since such a path exists for all $p, q \in A$, the annulus is path-connected.

Figure 11: Open annulus $A = \{1 < \|z\| < 2\}$ with a piecewise path $\gamma = \gamma_1 * \gamma_2 * \gamma_3$ connecting p to q inside A .

3.5 Connected but Not Path-Connected (Topologist's Sine Curve)

The set: Consider the set

$$S = \{(x, \sin(1/x)) : 0 < x \leq 1\} \cup (\{0\} \times [-1, 1]) \subset \mathbb{R}^2.$$

The first part oscillates infinitely often as $x \rightarrow 0^+$, and the vertical segment at $x = 0$ is included.

What the Picture is Conveying? The oscillating curve accumulates onto the entire segment $\{0\} \times [-1, 1]$: every point $(0, y)$ with $y \in [-1, 1]$ is a limit of points $(x_n, \sin(1/x_n))$ as $x_n \rightarrow 0^+$. So the set “sticks together” by accumulation, even though it does not provide a traversable corridor for paths.

Theorem 3.3 (Classical fact). *The set S is connected but not path-connected.*

Geometric explanation. (Connectedness). Let

$$C = \{(x, \sin(1/x)) : 0 < x \leq 1\}.$$

The map $x \mapsto (x, \sin(1/x))$ is continuous on $(0, 1]$, and $(0, 1]$ is connected, hence C is connected. Moreover, the closure $\overline{C}$ is connected (closure of a connected set in $\mathbb{R}^2$ is connected). One checks that

$$\overline{C} = C \cup (\{0\} \times [-1, 1]) = S.$$

Therefore S is connected.

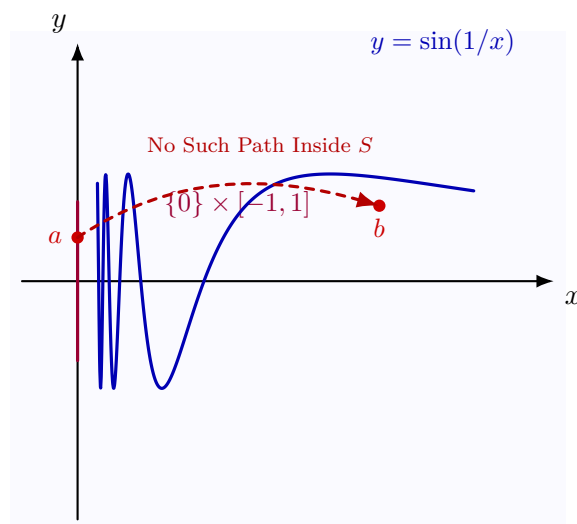


Figure 12: Topologist's sine curve (closure): a standard example of a set that is connected but not path-connected.

Not Path-connected: Intuitively, any continuous path starting at a point on the vertical segment $\{0\} \times [-1, 1]$ would have to move into $x > 0$ to reach the oscillating part, but near $x = 0$ the curve oscillates infinitely many times between $y = -1$ and $y = 1$ at arbitrarily small x , creating an obstruction: a path would have to “choose” a continuous route through infinitely many oscillations, which is impossible while staying inside S . Formally, one proves that there is no continuous $\gamma : [0, 1] \rightarrow S$ with $\gamma(0) \in \{0\} \times [-1, 1]$ and $\gamma(1) \in C$. Hence S is not path-connected. $\square$

Topologist's Sine Curve (Geometric Cue & Takeaway):

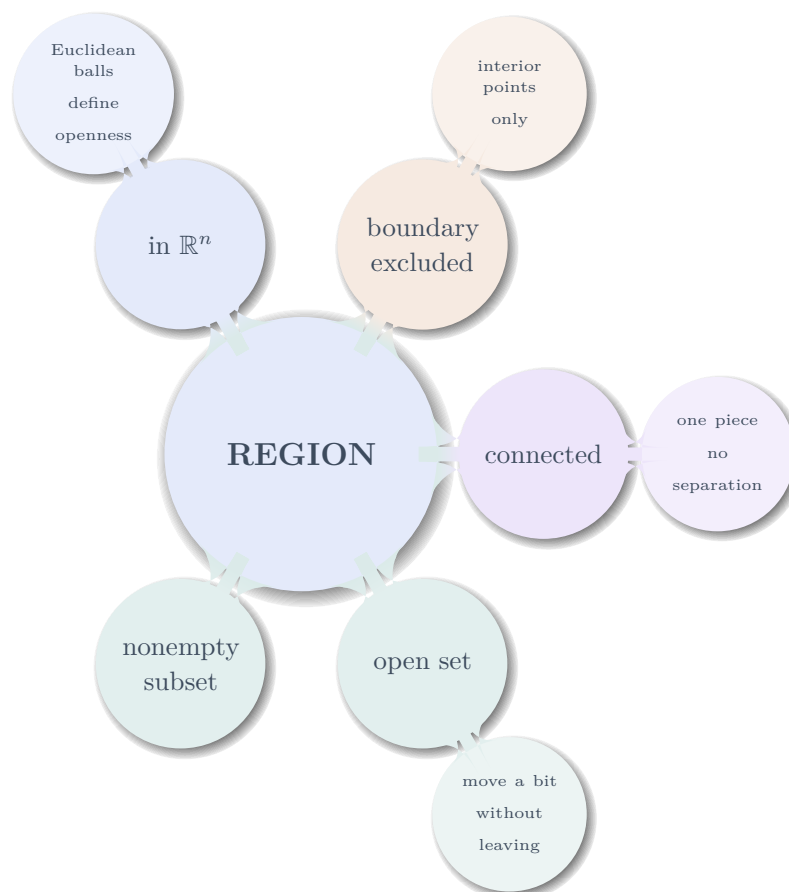
As $x \rightarrow 0^+$, the graph of $y = \sin(1/x)$ oscillates infinitely often, and its points accumulate onto the entire vertical segment $\{0\} \times [-1, 1]$. (**Accumulation Near $x = 0$**)

- **Key Takeaway:** This example shows the implication can fail: connected $\not\Rightarrow$ path-connected, even for subsets of $\mathbb{R}^2$.
- **Why Regions Behave Better:** For *open* subsets of $\mathbb{R}^n$ one has open and connected $\Rightarrow$ path-connected. Thus a *region* (a nonempty open connected set) is automatically path-connected, so connectedness and path-connectedness coincide for regions.

What the Collection of Diagrams Demonstrates

- **Open sets** are characterized by containing small balls around each of their points.
- **Disconnected sets** split into two separated, relatively open parts.
- **Path-connected sets** admit continuous curves joining any two points; hence they are connected.
- A **region** is a nonempty *open and connected* set; the annulus is a standard example.
- The **topologist's sine curve** shows that **connected** need not imply **path-connected**.

3.6 Mindmap Diagram



Common Pitfall. Clarification About “Doesn’t Include Boundary”

If U is open, then automatically $U \cap \partial U = \emptyset$. But a region may have a nonempty boundary as a *separate* set (e.g., open disk has circular boundary). What is excluded is that boundary points are not contained in the region.

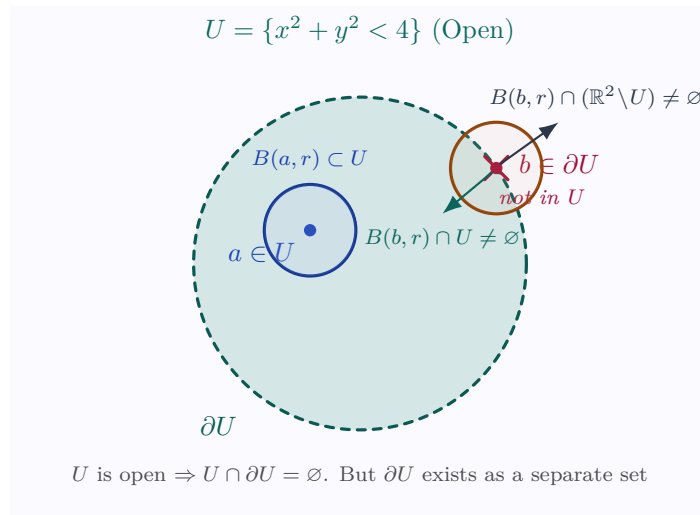


Figure 13: Open set vs. its boundary: boundary points exist, but they are excluded from the open set.

How to Read the Diagram (What It is Highlighting): The diagram is a visual proof of the statement

$$U \text{ open} \implies U \cap \partial U = \emptyset,$$

while also emphasizing that the boundary ∂U is a meaningful *separate* set.

(1) The Set U and Its Boundary: The filled teal region is the open disk

$$U = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < 4\}.$$

The dashed circle is the boundary

$$\partial U = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 4\}.$$

It is drawn dashed to indicate that these points are *not* part of the open set U .

(2) Interior Point: A Ball Stays Inside. A point $a \in U$ is marked inside the disk, together with a small circle $B(a, r)$. This shows the defining property of openness:

$$a \in U \implies \exists r > 0 \text{ such that } B(a, r) \subseteq U.$$

Geometrically: you can draw a small neighborhood around a that remains completely inside U .

(3) Boundary Point: Every Ball Meets Both Sides. A point b is marked on the dashed circle, so $b \in \partial U$. Around b we draw a small neighborhood $B(b, r)$ and indicate (with arrows) that it intersects:

$$B(b, r) \cap U \neq \emptyset \quad \text{and} \quad B(b, r) \cap (\mathbb{R}^2 \setminus U) \neq \emptyset.$$

This is exactly the neighborhood characterization of boundary points:

$$b \in \partial U \iff \forall r > 0, B(b, r) \cap U \neq \emptyset \text{ and } B(b, r) \cap (\mathbb{R}^2 \setminus U) \neq \emptyset.$$

So boundary points are precisely those points where you can never choose a small ball that lies entirely in U (or entirely in the complement).

(4) Why This Proves $U \cap \partial U = \emptyset$. If $b \in \partial U$, then *no* ball around b can be contained in U (because every ball also meets $\mathbb{R}^2 \setminus U$). But openness says: if a point lies in U , then *some* ball around it *is* contained in U . These conditions are incompatible, hence a boundary point cannot be an interior point:

$$b \in \partial U \Rightarrow b \notin U \implies U \cap \partial U = \emptyset.$$

(5) The Important Takeaway About “Regions”. Even though an open set (hence a region) excludes its boundary points, the boundary can still be nonempty. For the open disk, ∂U is the entire circle $\{x^2 + y^2 = 4\}$: it exists as a geometric object, but it is *not contained* in the region.

3.7 Theorem: Open Connected Implies Path-Connected in $\mathbb{R}^n$

Theorem. Open Connected Sets in $\mathbb{R}^n$ are Path-Connected

If $U \subseteq \mathbb{R}^n$ is open and connected, then U is path-connected.

Proof: Fix $x \in U$ and define

$$A := \{y \in U : \text{there exists a path in } U \text{ from } x \text{ to } y\}.$$

We show A is nonempty (since $x \in A$), open in U , and also closed in U .

(1) *A is open in U.* Let $y \in A$. Since U is open, there exists $r > 0$ with $B(y, r) \subseteq U$. For any $z \in B(y, r)$, the straight line segment

$$\gamma(t) = (1 - t)y + tz, \quad t \in [0, 1],$$

lies in $B(y, r) \subseteq U$. Concatenating the path from x to y with this segment gives a path from x to z . Hence $z \in A$, so $B(y, r) \cap U \subseteq A$. Therefore A is open in U .

(2) *$U \setminus A$ is open in U.* Take $y \in U \setminus A$. Again choose $r > 0$ such that $B(y, r) \subseteq U$. If $B(y, r)$ met A , say at $z \in A$, then the line segment in $B(y, r)$ would connect z to y inside U , and concatenating with a path from x to z would produce a path from x to y , contradicting $y \notin A$. So $B(y, r) \cap U \subseteq U \setminus A$, making $U \setminus A$ open in U .

Thus A is both open & closed in the connected space U , and $A \neq \emptyset$. Therefore $A = U$. Hence every $y \in U$ can be joined to x by a path; U is path-connected. $\square$

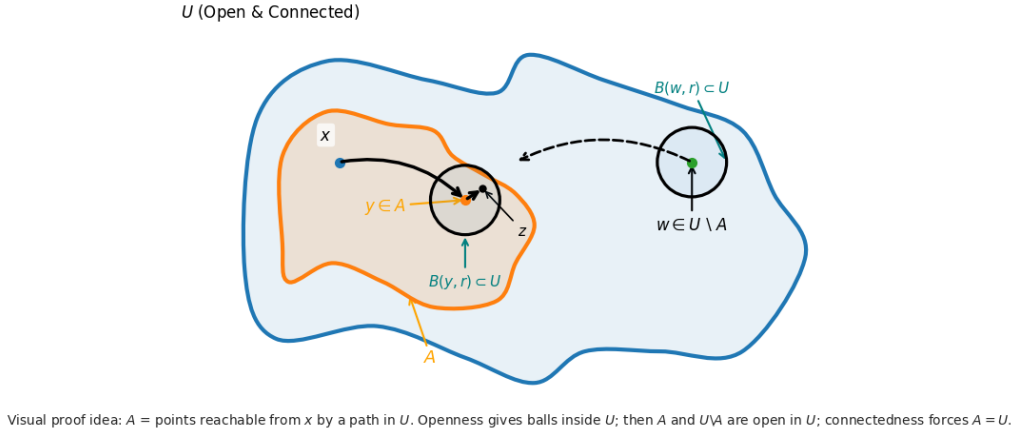


Figure 14: Visual roadmap of the proof that an open connected set $U \subset \mathbb{R}^n$ is path connected

3.8 Visual Proof (Guided by the Diagram)

Let $U \subseteq \mathbb{R}^n$ be **open and connected**. Fix a base point $x \in U$ (the point labeled x in the diagram). Define the *reachable set*

$$A := \{y \in U : \text{there exists a continuous path } \gamma : [0, 1] \rightarrow U \text{ with } \gamma(0) = x, \gamma(1) = y\}.$$

In the picture, U is the **large blue region** (open & connected), while A is the **orange subregion** inside U consisting of points you can reach from x by moving continuously while staying inside U .

Step 1: $A \neq \emptyset$. The constant path $\gamma(t) \equiv x$ lies in U and joins x to itself, so $x \in A$. Hence A is nonempty.

Step 2: A is open in U (the ball around y). Take any point $y \in A$ (the point labeled $y \in A$ in the diagram). Because U is *open*, there exists $r > 0$ such that the entire ball

$$B(y, r) \subset U$$

(the circle drawn around y) lies inside U .

Now choose any point $z \in B(y, r)$ (the nearby point labeled z). Since Euclidean balls are convex, the straight segment

$$\sigma(t) = (1 - t)y + tz, \quad t \in [0, 1],$$

stays inside $B(y, r)$, hence inside U .

Because $y \in A$, there is already a path γ in U from x to y (the thick curved arrow from x into the orange region). Concatenating γ with the segment σ yields a continuous path in U from x to z . Therefore $z \in A$.

Since *every* $z \in B(y, r)$ lies in A , we have

$$B(y, r) \cap U = B(y, r) \subseteq A,$$

so A is open in the subspace topology on U .

Step 3: $U \setminus A$ is open in U (the ball around w). Take a point $w \in U \setminus A$ (the point labeled $w \in U \setminus A$ in the diagram). Again, openness of U gives an $r > 0$ such that

$$B(w, r) \subset U$$

(the circle drawn around w).

We claim $B(w, r)$ cannot intersect A . If it did, pick $z \in B(w, r) \cap A$. Then there is a path in U from x to z (because $z \in A$). Also, since $z, w \in B(w, r)$, the straight segment from z to w lies entirely inside the ball $B(w, r) \subset U$. Concatenating these two paths produces a path in U from x to w , which would imply $w \in A$ —a contradiction.

Hence

$$B(w, r) \cap A = \emptyset \implies B(w, r) \cap U = B(w, r) \subseteq U \setminus A,$$

so $U \setminus A$ is open in U .

Step 4: Connectedness forces $A = U$. We have written U as a disjoint union

$$U = A \cup (U \setminus A),$$

and Steps 2–3 show that both A and $U \setminus A$ are open in U . Since $A \neq \emptyset$ and U is *connected*, the only possibility is

$$U \setminus A = \emptyset \implies A = U.$$

Conclusion (Path-Connectedness): Because $A = U$, every point of U can be joined to x by a path in U . In particular, given any $p, q \in U$, apply the above argument with $x = p$. Then $q \in A = U$, so there exists a path in U from p to q . Thus U is **path-connected**. ■

3.9 Solved Examples

Solved Example. Open Disk is a Region; Closed Disk is Not

Let

$$U = \{x \in \mathbb{R}^2 : \|x\| < 1\}, \quad D = \{x \in \mathbb{R}^2 : \|x\| \leq 1\}.$$

Show that U is a region, but D is not a region.

Solution: A *region* in $\mathbb{R}^n$ means: **nonempty + open + connected**. So we check these properties for U and D . **Part A: U is a Region.**

- **Nonempty:** Clearly $0 \in U$ since $\|0\| = 0 < 1$.
- **U is open:** Take any $x \in U$. Then $\|x\| < 1$, so the number

$$r := 1 - \|x\| > 0$$

is strictly positive. We claim $B(x, r) \subset U$. Indeed, if $y \in B(x, r)$, then $\|y - x\| < r$, hence by the triangle inequality,

$$\|y\| \leq \|y - x\| + \|x\| < r + \|x\| = (1 - \|x\|) + \|x\| = 1.$$

So $\|y\| < 1$, which means $y \in U$. Therefore $B(x, r) \subset U$ and U is open.

- **U is connected (In fact Path-Connected):** A very strong sufficient condition is **convexity**. We show U is convex: take $p, q \in U$ and $t \in [0, 1]$. Consider

$$\gamma(t) = (1 - t)p + tq.$$

$$\implies \|\gamma(t)\| = \|(1 - t)p + tq\| \leq (1 - t)\|p\| + t\|q\| < (1 - t) \cdot 1 + t \cdot 1 = 1,$$

since $\|p\| < 1$ and $\|q\| < 1$. Hence $\gamma(t) \in U$ for all t , giving a path inside U from p to q . So U is path-connected. Hence, U is connected.

Conclusion for U is nonempty, open, and connected, hence U is a region.

Part B: D is not a Region:

- **D is Nonempty & Connected:** It is nonempty since $0 \in D$. It is also convex (same argument as above but using ≤ 1), so it is path-connected and hence connected.
- **D is Not Open:** To show a set is not open, it suffices to find *one point* that has *no ball contained in the set*. Pick a boundary point, e.g.

$$x = (1, 0) \in D, \quad \|x\| = 1.$$

Let $r > 0$ be any radius. Consider the point

$$y = (1 + \frac{r}{2}, 0).$$

Then $\|y - x\| = \frac{r}{2} < r$, so $y \in B(x, r)$, but $\|y\| = 1 + \frac{r}{2} > 1$, hence $y \notin D$.

So, no matter how small $r > 0$ is, $B(x, r) \not\subseteq D$. Therefore D is not open. Since D fails to be open, it cannot be a region.

Common Pitfall. Key Lesson

The difference between U and D is **boundary inclusion**. Open sets cannot contain their boundary points (indeed $U \cap \partial U = \emptyset$), but closed sets often *do* contain boundary points, which is why they usually fail to be open.

Solved Example. A Set Can be Open But Not Connected

Let

$$E = (-2, -1) \cup (1, 2) \subseteq \mathbb{R}.$$

Decide whether E is a region in $\mathbb{R}$.

Solution:

Step 1: Check Openness. Both $(-2, -1)$ and $(1, 2)$ are open intervals in $\mathbb{R}$, hence open sets. A union of open sets is open, so

$$E = (-2, -1) \cup (1, 2)$$

is open.

Step 2: Check Connectedness: Intuitively E has a gap around 0. Formally, define

$$U := (-2, -1), \quad V := (1, 2).$$

Then U and V are nonempty, disjoint, open in $\mathbb{R}$, and

$$E = U \cup V.$$

Moreover, U and V are also open in the *subspace topology on E* :

$$U = E \cap (-2, -1), \quad V = E \cap (1, 2),$$

with $(-2, -1)$ and $(1, 2)$ open in $\mathbb{R}$. Thus E is disconnected.

Conclusion: A region in $\mathbb{R}$ must be open and connected. Although E is open, it is not connected. Hence E is **not** a region in $\mathbb{R}$.

3.10 Exercises

Exercises. Orbits: Bounded vs Escaping vs Periodic

1. For $f(x) = 2x$ on $\mathbb{R}$, determine $\lim_{n \rightarrow \infty} f^n(x)$ for $x = 0$ and for $x \neq 0$.
2. For $f(x) = \frac{1}{2}x$ on $\mathbb{R}$, show every orbit converges to 0.
3. Prove: x is periodic of period p iff the orbit $\{f^n(x)\}$ has exactly p distinct points.
4. For $f_c(z) = z^2 + c$, prove the escape lemma for the simpler radius $R = 2$ under the assumption $|c| \leq 2$.
5. Let R_θ be rotation on $\mathbb{T}$. Prove: if $\theta/2\pi$ is rational, every orbit is periodic; if irrational, no orbit is periodic.

3.11 Additional Exercises on $f_c(z) = z^2 + c$ (Bounded vs Escaping)

Exercises. E1: Orbit table for a real parameter

Let $c = -\frac{3}{4}$ and $z_0 = 0$. Compute $z_1, \dots, z_8$ (as real numbers) and decide whether the orbit appears bounded.

Exercises. E2: Detect a 2-cycle

Let $c = -1$ and $z_0 = 1$. Compute z_1, z_2, z_3, z_4 and determine whether the orbit falls into a periodic cycle.

Exercises. E3: Fixed points for a given c

Let $c = -\frac{1}{4}$. Solve $z = f_c(z)$ explicitly (find all fixed points). For each fixed point z^* compute $f'_c(z^*)$ and decide whether the fixed point is attracting, repelling, or neutral.

Exercises. E4: Escape test with a large real seed

Let $c = -0.2$ and $z_0 = 3$. Use the escape radius 2 to prove the orbit escapes without computing many iterates.

Exercises. E5: First escape time

Let $c = 0$ and $z_0 = 1.2$. Compute the smallest n such that $|z_n| > 2$ (if it happens), and hence conclude escape.

Exercises. E6: Complex arithmetic practice

Let $c = 0.2 + 0.3i$ and $z_0 = 0$. Compute z_1, z_2, z_3 explicitly in $a + bi$ form and compute $|z_1|, |z_2|, |z_3|$.

Exercises. E7: Compare two seeds for the same c

Fix $c = -0.12 + 0.74i$. Compute the first 6 iterates for $z_0 = 0$ and for $w_0 = 1$. Which orbit appears closer to escaping?

Exercises. E8: Escape criterion proof (core lemma)

Assume $|c| \leq 2$. Prove that if $|z_n| > 2$ for some n , then $|z_{n+k}| \rightarrow \infty$ as $k \rightarrow \infty$. (Hint: show $|z_{n+1}| \geq |z_n|^2 - 2$.)

Exercises. E9: An improved escape radius

Show that for any $c \in \mathbb{C}$, if $|z_n| > R := \max\{2, |c| + 1\}$ for some n , then the orbit escapes.

Exercises. E10: Boundedness of a constant orbit

Find all $c \in \mathbb{C}$ such that $z_0 = 0$ produces the constant orbit $z_n \equiv 0$ for all n .

Exercises. E11: Preperiodic example

Let $c = -1$ and $z_0 = 0$. Show that 0 is periodic. Now start at $z_0 = -1$ and show it is also periodic. Finally, start at $z_0 = 1$ and determine whether it is periodic or preperiodic.

Exercises. E12: Real dynamics window

Let $c = -2$ and $z_0 = 0$. Compute z_1, z_2, z_3, z_4 and decide whether the orbit escapes. Then try $z_0 = 0.1$ and compare.

Exercises. E13: Solve for c from a desired fixed point

Let $a \in \mathbb{C}$ be given. Find all $c \in \mathbb{C}$ such that a is a fixed point of f_c , i.e. $f_c(a) = a$.

Exercises. E14: Solve for c from a desired 2-cycle

Let $p, q \in \mathbb{C}$ with $p \neq q$. Find the condition on c such that $p \mapsto q \mapsto p$ is a 2-cycle of f_c . (Write equations relating p, q, c .)

Exercises. E15: Multiplier of a periodic orbit

Let z^* be a fixed point of f_c . Show that the multiplier is $f'_c(z^*) = 2z^*$. More generally, if $(z_0, z_1, \dots, z_{p-1})$ is a p -cycle, show the cycle multiplier equals

$$\lambda = (f_c^p)'(z_0) = \prod_{k=0}^{p-1} 2z_k.$$

Exercises. E16: Basin-of-infinity check (numeric)

Let $c = 0.3 + 0.5i$. Choose seeds $z_0 \in \{0, 0.5, 1, 1+i, 2\}$ and iterate up to $N = 50$. Which seeds escape (cross $|z| > 2$) within 50 steps?

Exercises. E17: Sensitivity near the boundary

Fix $c = 0.3 + 0.5i$. Compare two nearby seeds $z_0 = 0.20 + 0.10i$ and $w_0 = 0.2001 + 0.10i$. Iterate both up to $N = 200$. Record the first n where $|z_n - w_n| > 10^{-2}$ (if it happens).

Exercises. E18: Real-only invariance

Assume $c \in \mathbb{R}$ and $z_0 \in \mathbb{R}$. Prove that all iterates z_n remain real. Then explain why the complex plane viewpoint still matters for Julia sets.

Exercises. E19: Conjugacy symmetry

Let $c \in \mathbb{C}$ and define $\overline{f_c(z)} = \overline{z^2 + c}$. Show that

$$\overline{f_c(z)} = f_{\bar{c}}(\bar{z}).$$

Deduce that the dynamics for c and $\bar{c}$ are mirror images across the real axis.

Exercises. E20: Forward-invariant disk (simple trapping region)

Let $|c| \leq \frac{1}{4}$. Show that the closed disk $\{z : |z| \leq \frac{1}{2}\}$ is forward invariant under f_c (i.e. $|z| \leq \frac{1}{2} \Rightarrow |f_c(z)| \leq \frac{1}{2}$). Conclude that seeds in this disk have bounded orbits.

Exercises. E21: Show escape for large $|c|$ at the critical point

Let $z_0 = 0$ and suppose $|c| > 2$. Show that the critical orbit escapes (hint: $z_1 = c$ already satisfies $|z_1| > 2$).

Exercises. E22: Orbit portrait (hand sketch + computation)

Fix $c = -1$ and compute the orbit starting at $z_0 = i$ for $n = 0, 1, 2, 3, 4, 5$. Plot the points $(\Re z_n, \Im z_n)$ by hand on the complex plane and connect consecutive iterates with arrows. Does the orbit appear bounded?

Exercises. E23: Escape-time function (definition + computation)

For fixed c , define the escape time

$$T(z_0) = \min\{n \geq 0 : |z_n| > 2\}, \quad \text{with } T(z_0) = \infty \text{ if no such } n.$$

For $c = 0$, compute $T(z_0)$ for $z_0 \in \{0, 0.5, 1.1, 1.5, 2.1\}$.

Exercises. E24: Compare two parameters

Take $c_1 = -1$ and $c_2 = 0.3 + 0.5i$, with seed $z_0 = 0$ in both cases. Compute the first 6 iterates in each case and compare boundedness/escape behavior.

3.12 Exercises**Exercises. Boundary Practice**

1. Compute $\partial(2, 5)$, $\partial[2, 5]$, $\partial(2, 5]$ in $\mathbb{R}$.
2. In $\mathbb{R}^2$, find the boundary of an open disk $B(0, 1)$ and the closed disk $\overline{B}(0, 1)$.
3. Prove $\partial A = \overline{A} \cap (\overline{X \setminus A})$ directly from the neighborhood definition.
4. Show: A is open iff $A \cap \partial A = \emptyset$.
5. Show: $\partial(A \cup B) \subseteq (\partial A \cup \partial B)$ and give an example where strict inclusion holds.

3.13 Exercises**Exercises. Regions**

1. Show that any open interval $(a, b) \subseteq \mathbb{R}$ is a region.
2. Give an example of a connected set in $\mathbb{R}^2$ that is not path-connected.
3. Prove that if $U \subseteq \mathbb{R}^n$ is convex, then it is path-connected.
4. Let U be open. Prove $U \cap \partial U = \emptyset$.
5. Prove that if U is a region and $f : U \rightarrow \mathbb{R}$ is continuous, then $f(U)$ is an interval if $n = 1$ (Intermediate Value viewpoint).

4 Computational Explorations

A number of the diagrams and “numeric boundary tests” in these notes were also explored computationally in Python. The goal of the code is not to replace proofs, but to *support intuition* and to help you experiment with:

- Neighborhood tests for boundary/interior/exterior points in $\mathbb{R}$ and $\mathbb{R}^2$,
- Region intuition via open balls inside open sets,
- Orbit iteration (rotation on $\mathbb{T}$ and quadratic maps $z^2 + c$),
- Escape-time experiments that visually separate bounded vs. escaping behavior.

The complete Python notebooks/scripts and the generated figures are available on GitHub:

Access the GitHub repository for the Python visualizations

Remark 4.1. If you are reading a PDF printout, type the URL shown above into your browser. The repository contains reproducible code so you can modify parameters (radii, seeds, iteration counts, and sets A) and immediately see how the pictures change.

5 Conclusion

In these notes we built a single, consistent viewpoint: *local neighborhoods control global structure*. We began with boundary as the place where every neighborhood must “see both sides” (inside and outside a set). This naturally led to the closure/interior identities

$$\partial A = \overline{A} \setminus A^\circ \quad \text{and} \quad \partial A = \overline{A} \cap \overline{(X \setminus A)},$$

which convert geometric intuition into precise membership tests.

Next we shifted to *orbits under iteration* to connect repeated application of a map with long-term behavior: fixed points, periodic cycles, bounded motion, and escape. The rotation examples on the circle showed a clean dichotomy between rational and irrational angles, while the quadratic family $f_c(z) = z^2 + c$ illustrated a practical “escape-radius” idea that becomes the computational backbone of Julia and Mandelbrot explorations.

Finally, we studied *regions* in $\mathbb{R}^n$ (nonempty open connected sets) and proved the key analytic fact:

$$\text{open and connected in } \mathbb{R}^n \implies \text{path-connected.}$$

The proof was presented both in formal steps and in a diagram-guided visual argument, emphasizing how openness produces small balls and connectedness rules out a nontrivial clopen split. Along the way, the topologist’s sine curve reminded us that connectedness alone does not guarantee the existence of paths—highlighting why regions behave better in Euclidean analysis.

Takeaway. Whenever you feel stuck, return to the neighborhood principle: *test membership by balls, and test separation by whether balls can isolate pieces*. This mindset makes theorems feel inevitable rather than memorized.